# Bidirectional Astrocyte Neuron Signalling

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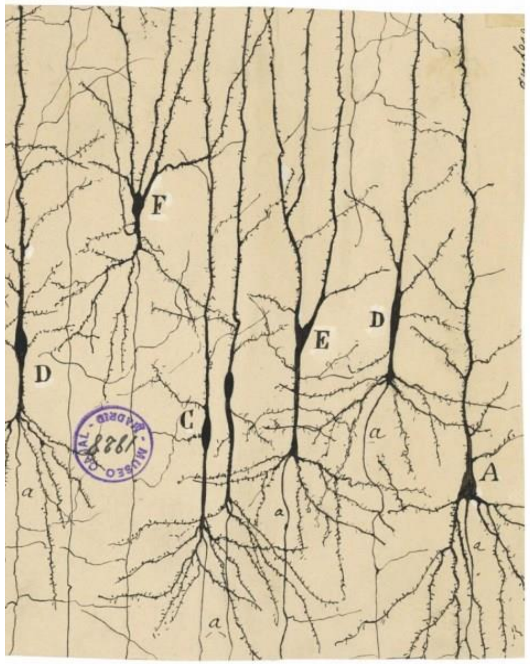
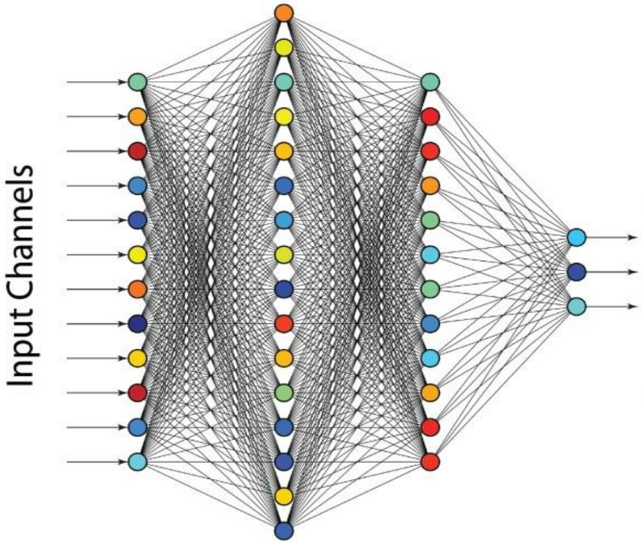
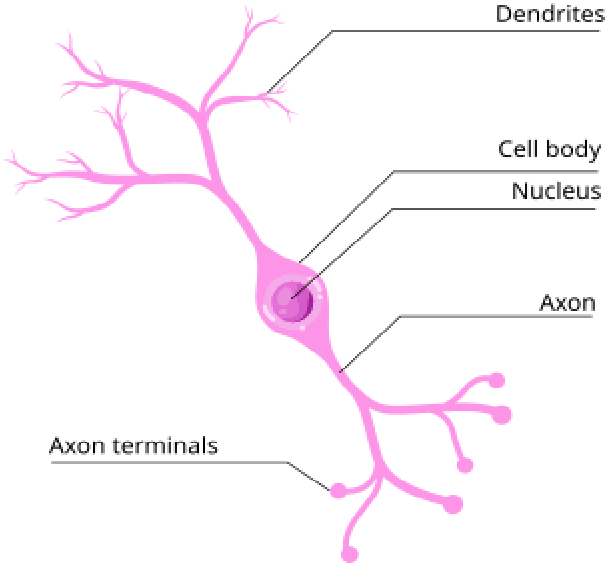
# Neural Networks: From Cajal to Hinton



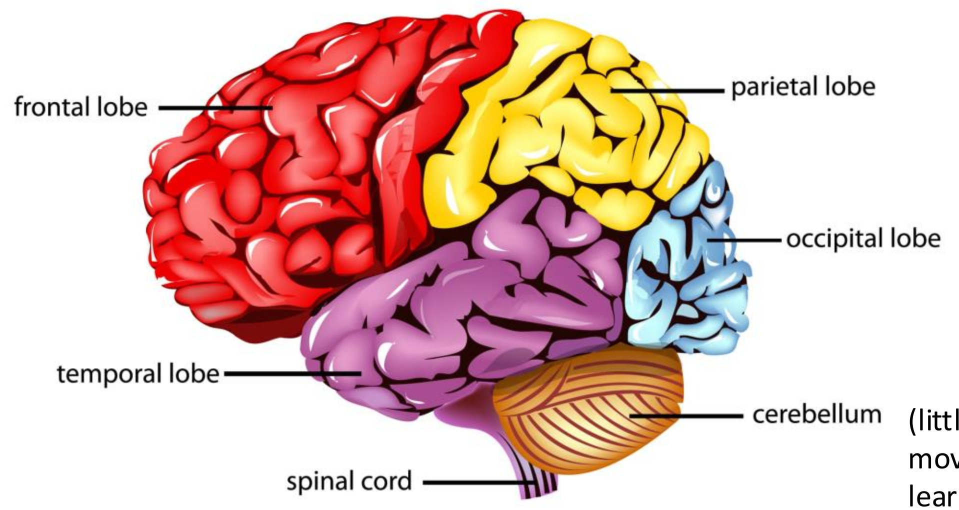




- Cajal used staining to distinguish neurons from neural matter - became known as the neuron doctrine - won Nobel Nobel Prize in physiology in 1906
- In 1986, Rumelhart, Hinton and Williams developed a mathematical procedure for training neural weights called error backpropagation

# Brain Regions

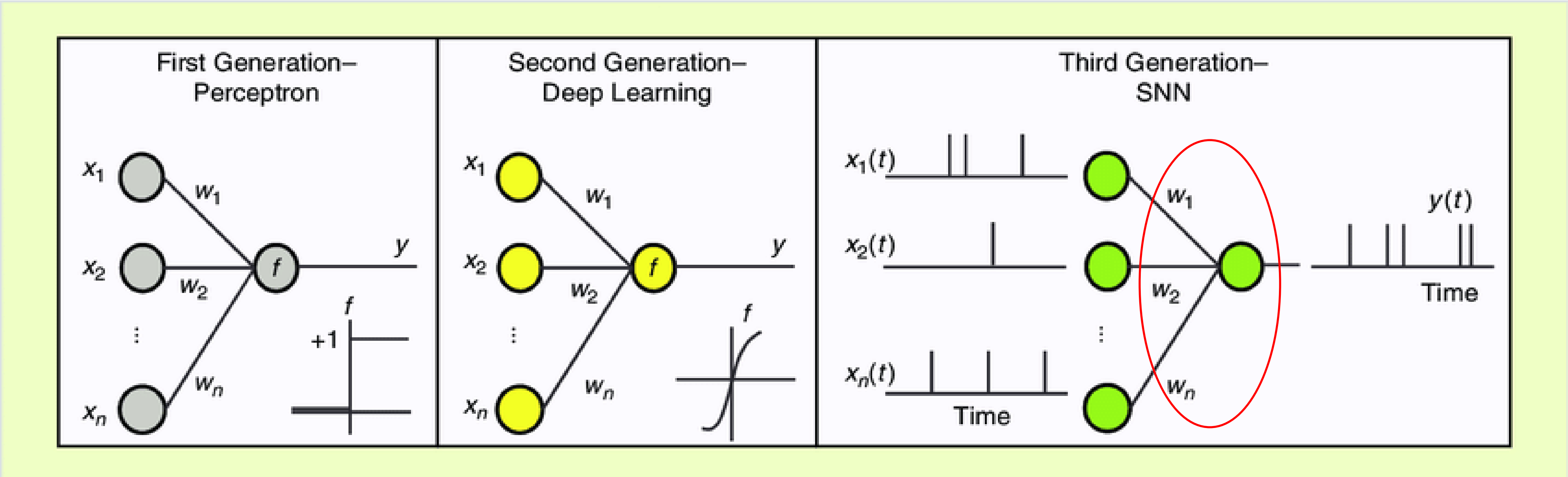
Human brain contains 86 billion neurons – half in cerebellum

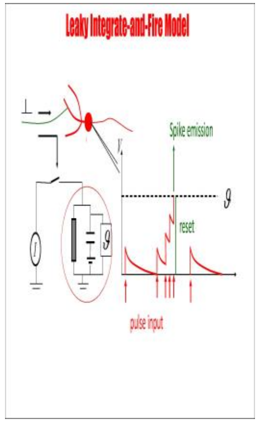



(little brain) - coordinates movements, balance, motor learning

- Frontal lobe - Behaviour and emotional control centre
- Parietal lobe - Processes sensory information from the outside world: touch, taste, and temperature
- Occipital lobe - Primarily responsible for visual processing
- Temporal lobe - Process emotions, language, and certain aspects of visual perception

# Generations of Neural Networks



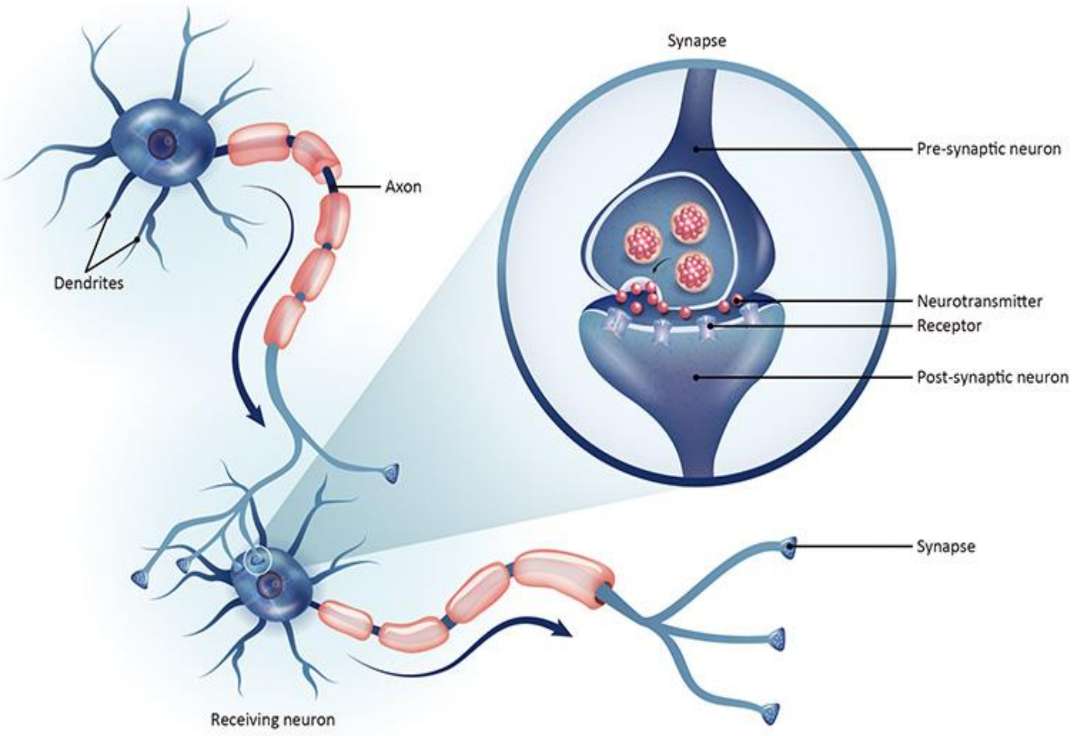



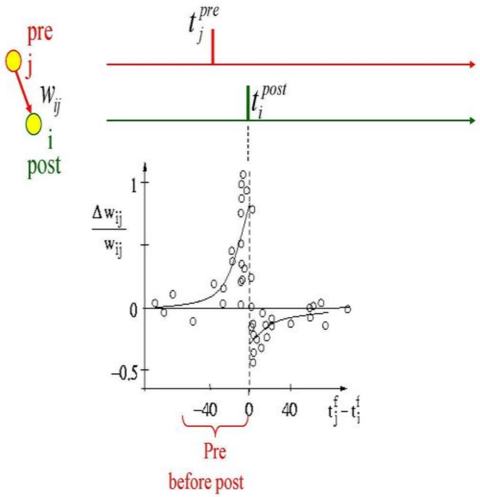

Perceptron - supervised image classification developed by Frank Rosenblatt in 1958 - could solve linearly separable problems

Sigmoidal - differentiable curve became foundational building block of ANN in the 1980s

SNNs evolved from decades of biological research – IF model in 1907 & H&H model in 1952 - Defined as 3rd generation by Maass in a 1997 paper

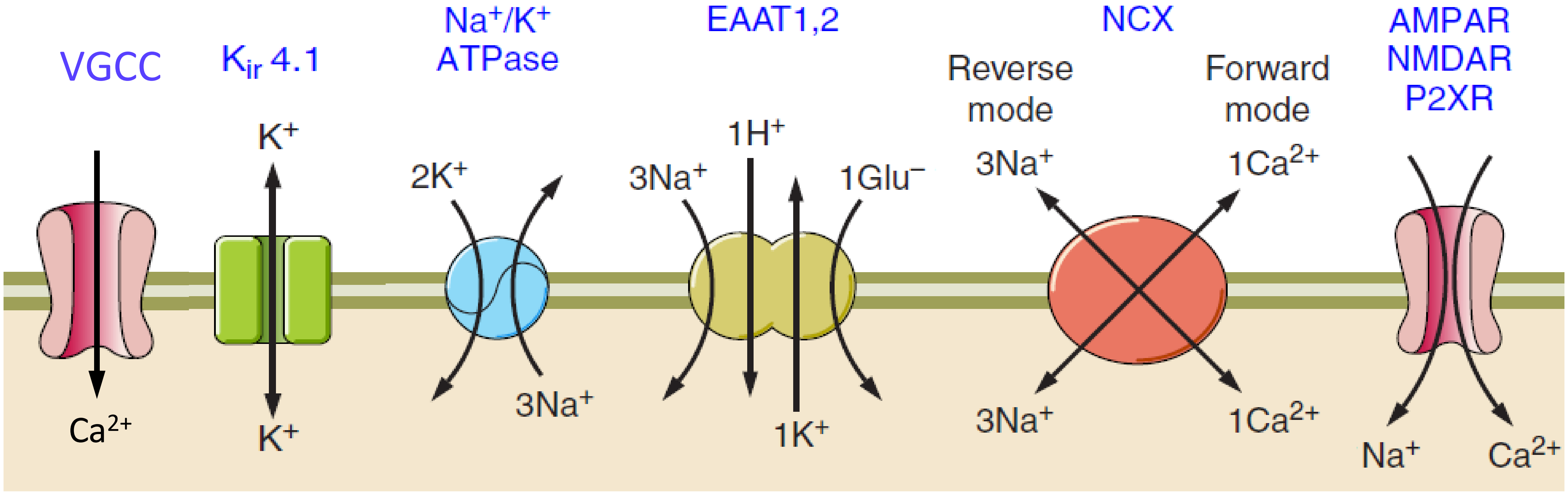
# Synaptic Junction between two Neurons



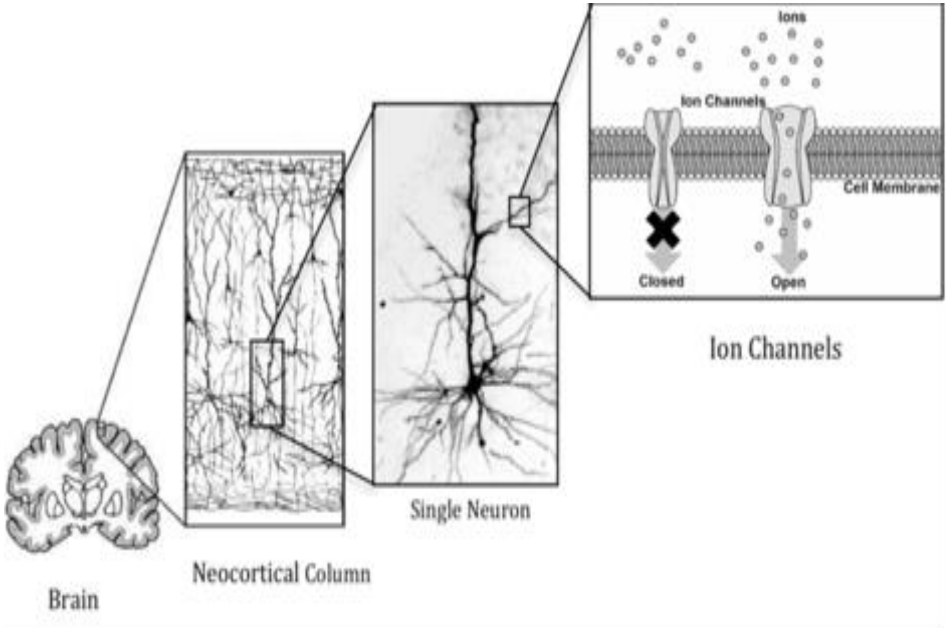

Spike Timing Dependent Plasticity (STDP) rule

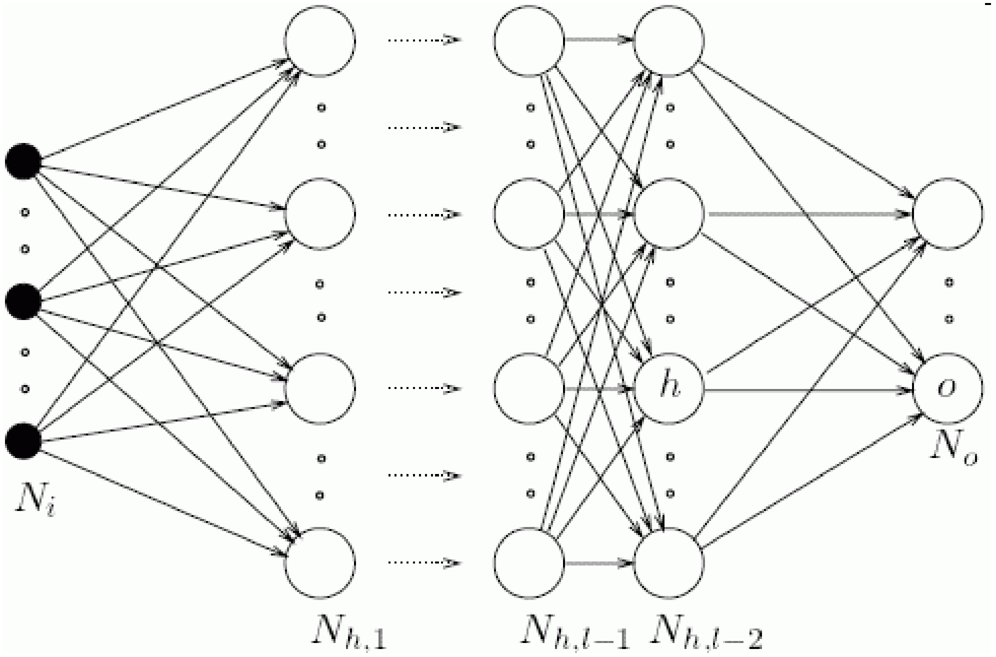
# Ionic Transport Across Cell Membranes




Reference: **Verkhratsky A, Nedergaard M.** Physiology of Astroglia. *Physiol Rev* 98: 239–389, 2018. doi:10.1152/physrev.00042.2016.

# Abstraction Level



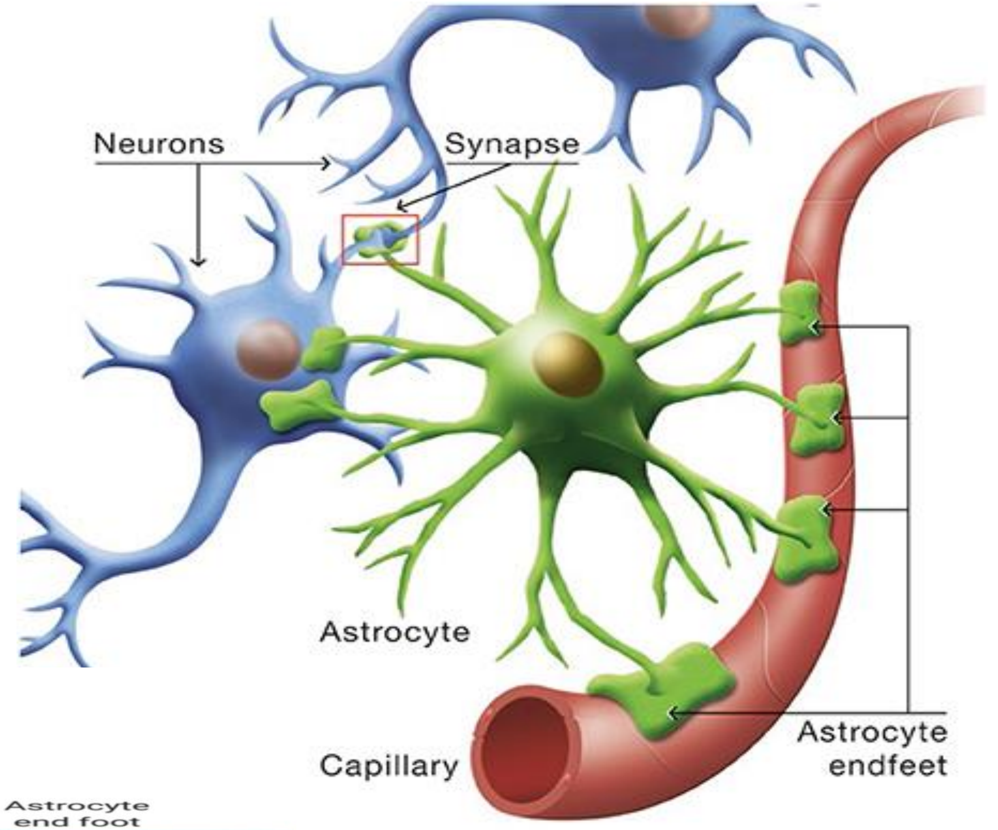

They can be trained to recognise patterns
Generalise when given unseen inputs

# New Player-Astrocyte

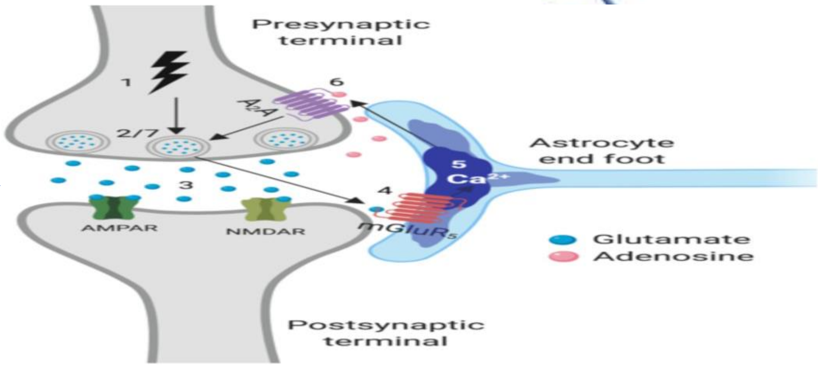
- Astrocytes (Astroglia) - star-shaped – astro (Greek for star) & glia (Greek for glue)
- Historically dismissed as passive structural scaffolding
- Occupy 25% to 50% of total brain volume - outnumbering neurons
- Astrocytes - type of glial cell that enwraps neuronal synapses - semi-isolate synapses
- Homeostatic control of different ionic species like $Ca^{2+}$, $Na^{+}$ and $K^{+}$
- Thin processes (∽100nm) play a crucial

  role in ionic flow – effectively semi-isolate

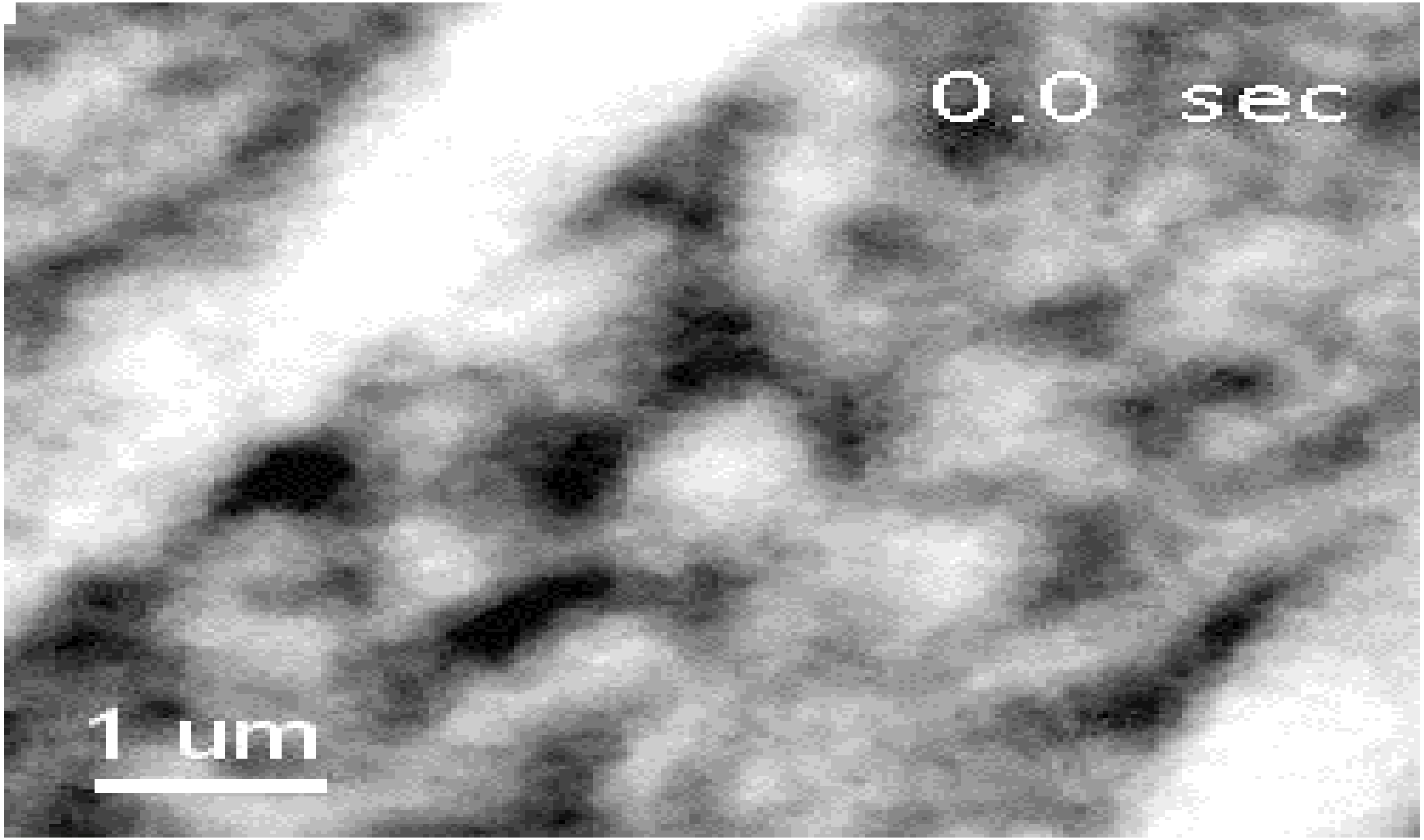
  cradle from soma



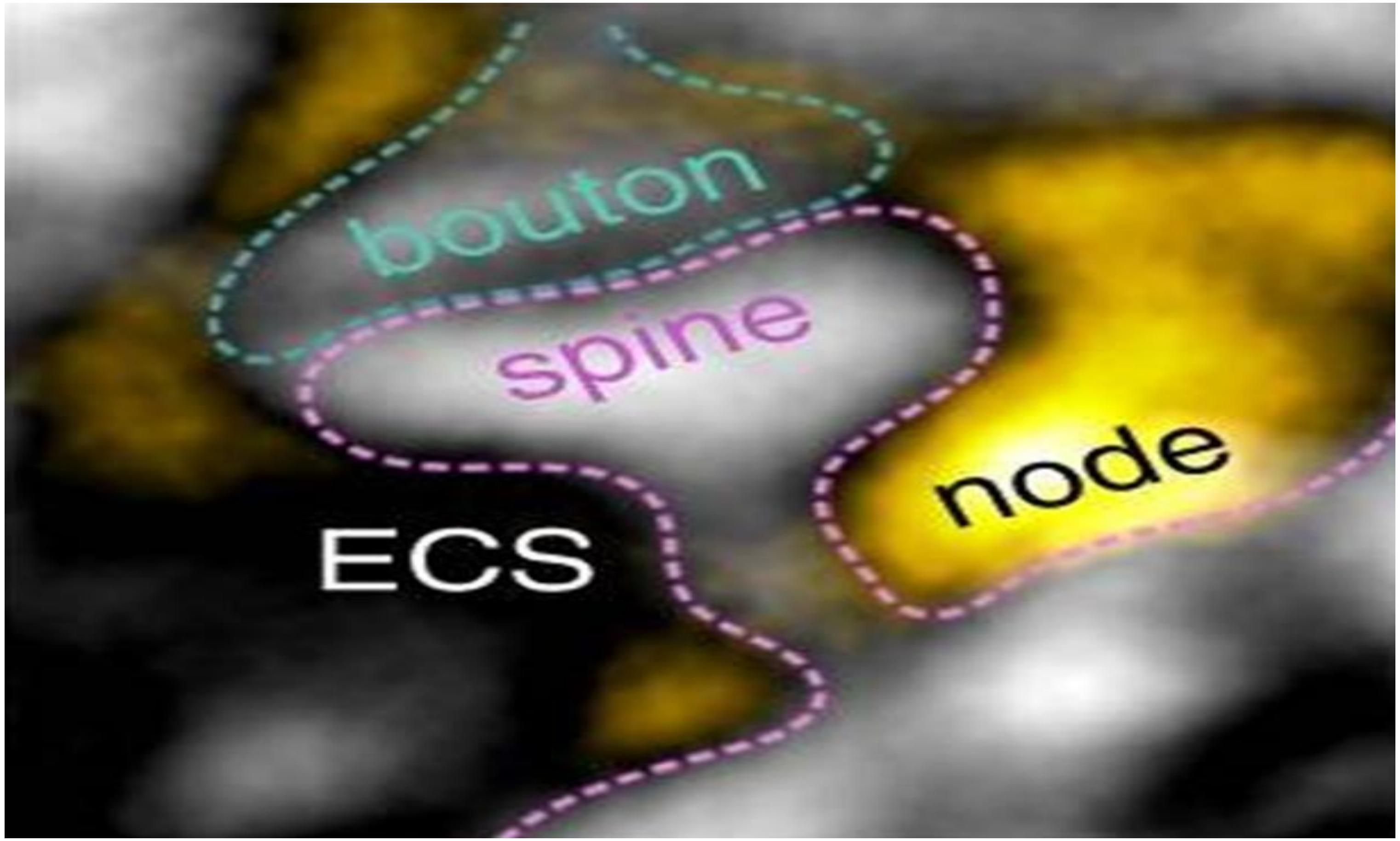

Tripartite synapse



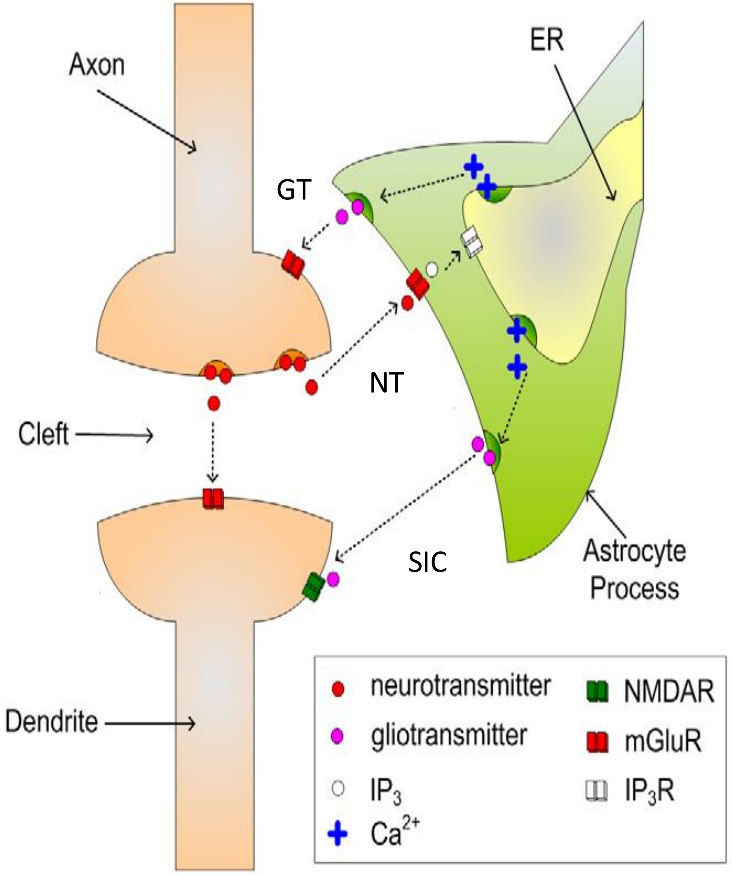
# Calcium Wave

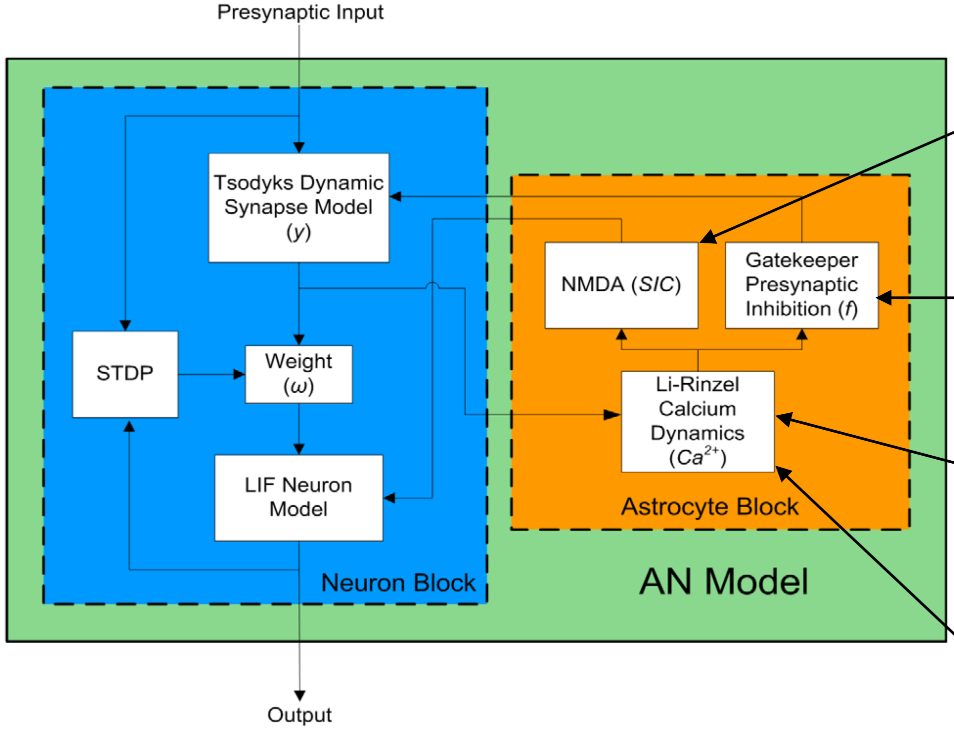


# Astrocyte Coverage at Tripartite Synapses



# Astrocyte-Neuron (AN) Model




$$\tau_{eSP}\frac{dSIC}{dt} = -SIC + mAS(t)$$

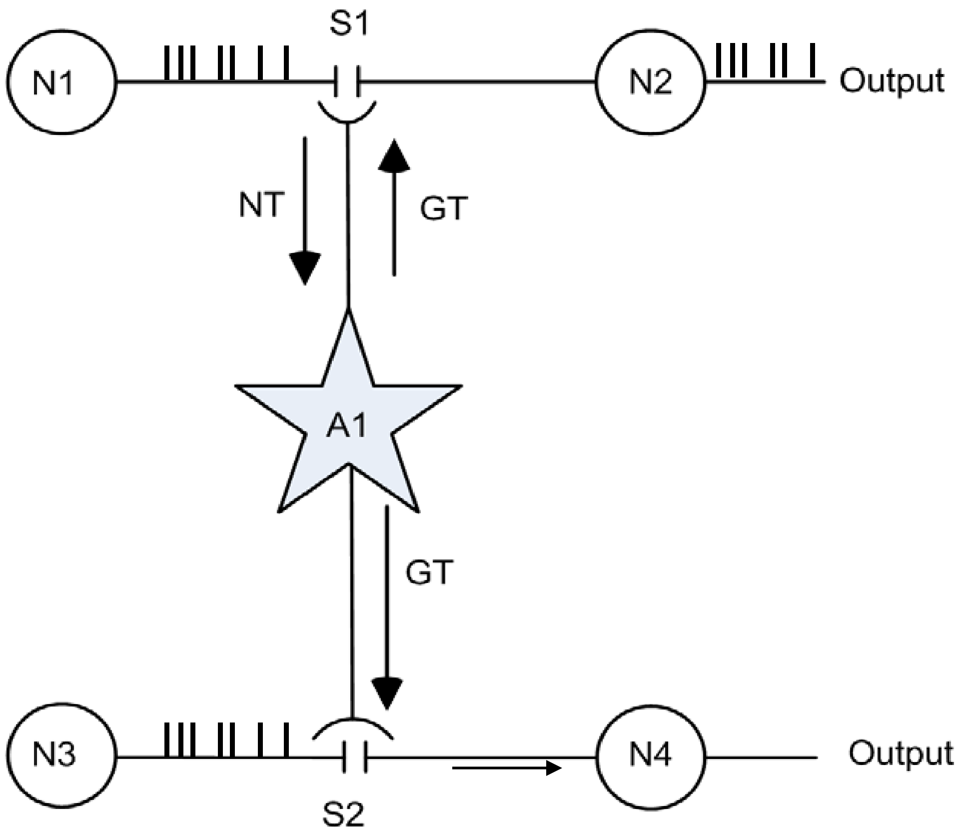
$$\frac{df}{dt} = \frac{-f}{\tau_{Ca2+}} + (1-f)k\Theta(\mathrm{Ca}^{2+} - Ca^{2+}_{thresh}l)$$

$$\frac{dIP_3}{dt} = \frac{IP_3^* - IP_3}{\tau_{ip3}} + r_{ip3}y$$

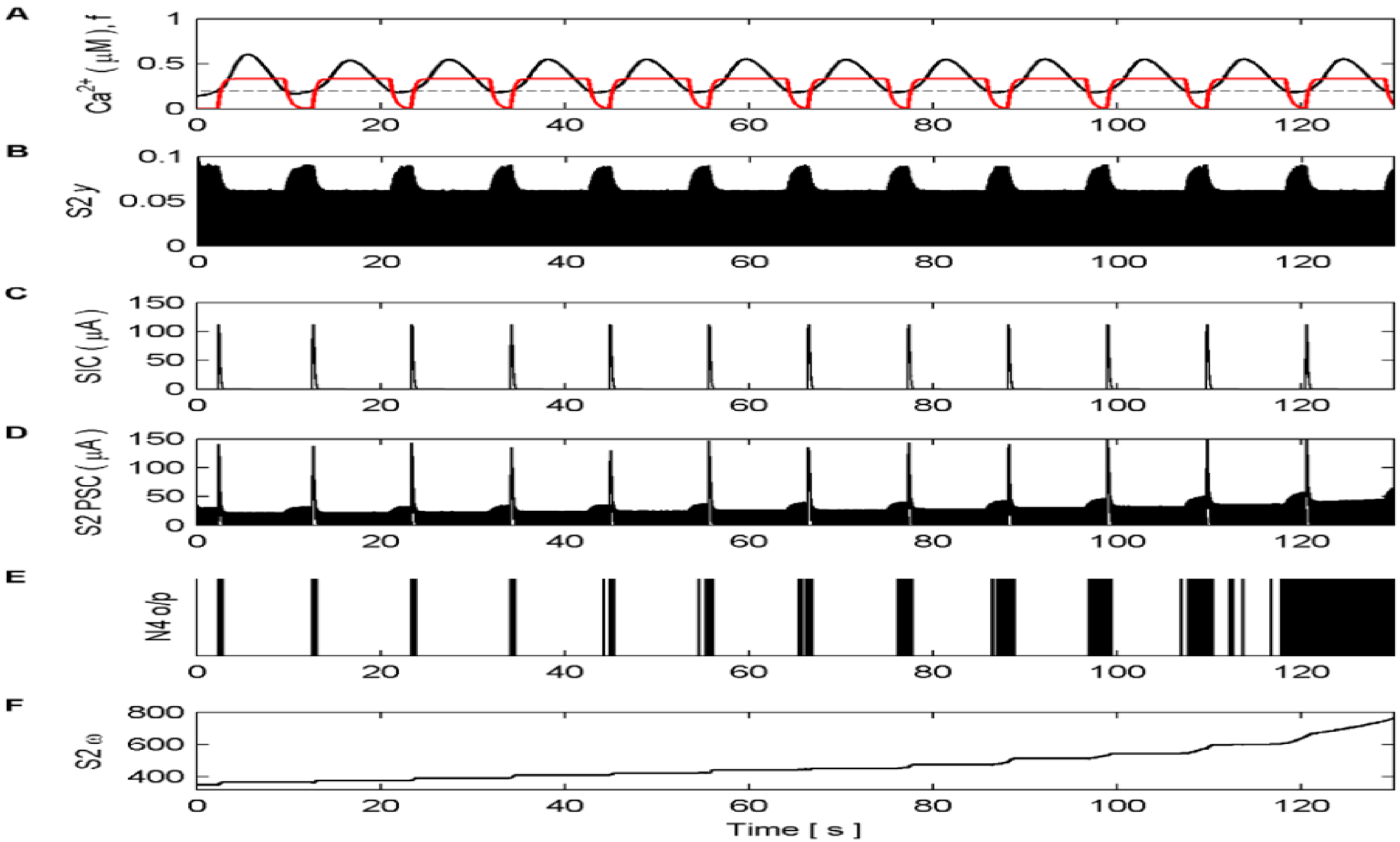
$$\frac{dCa^{2+}}{dt} = -J_{channel}(IP_3) - J_{pump} - J_{leak}$$

*f* is a phenomenological gating variable

# Multisynaptic Coupling for STDP

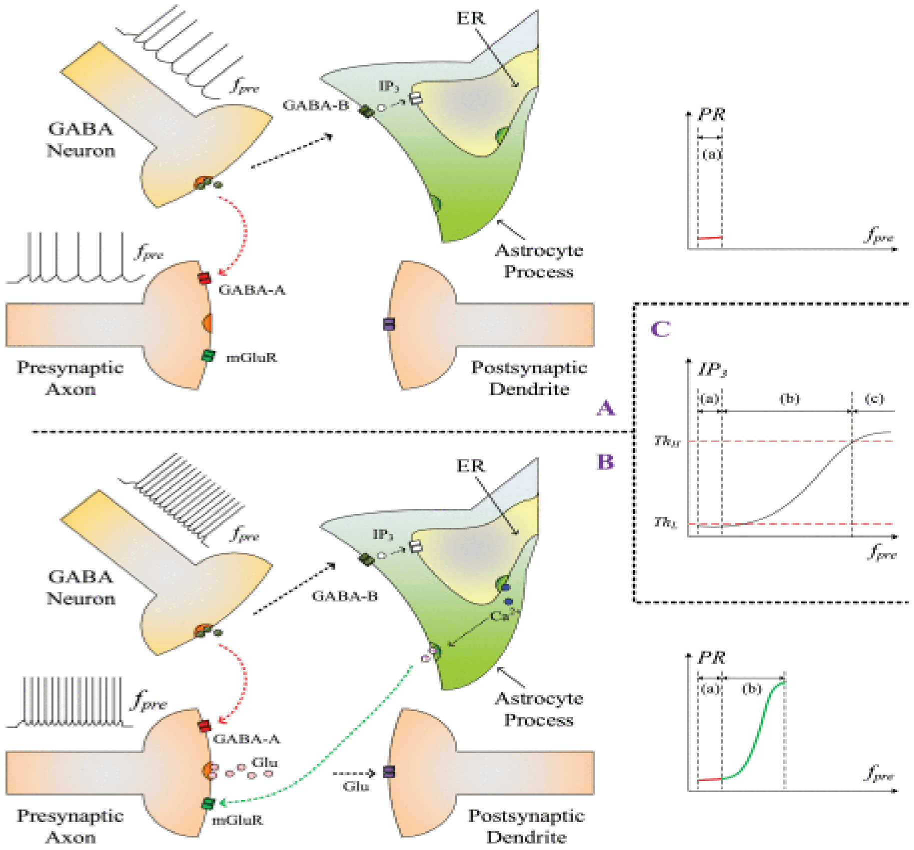


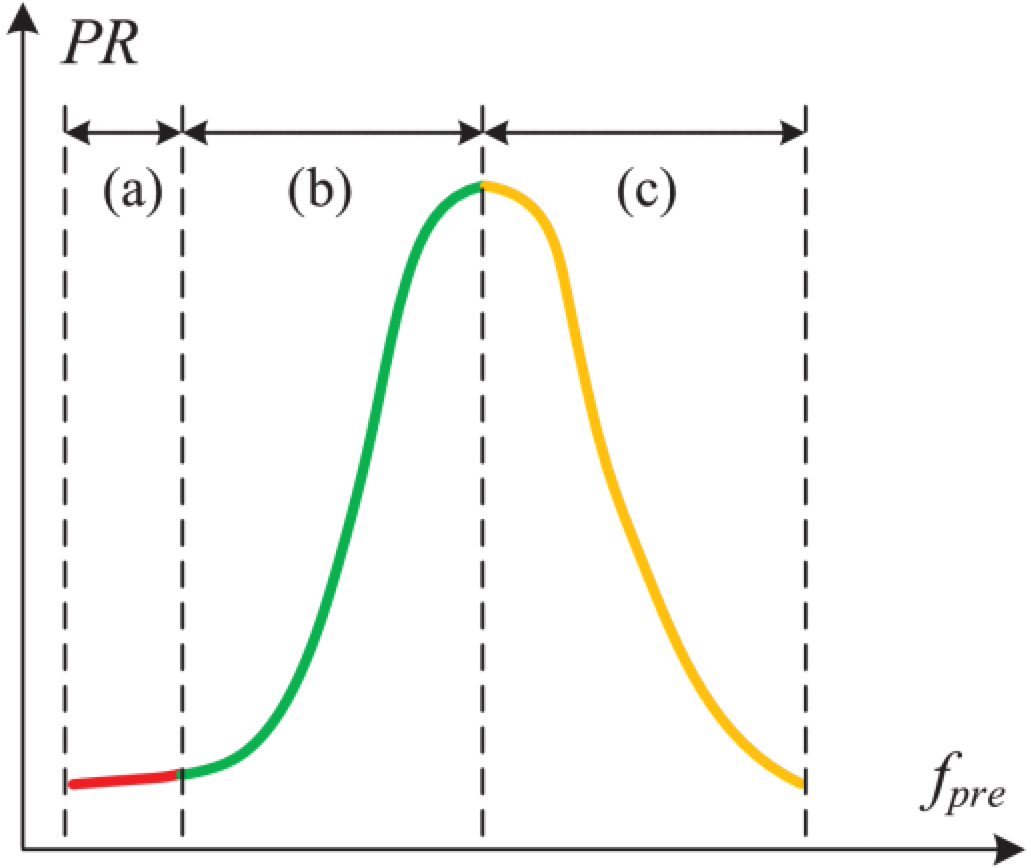
# Multisynaptic coupling for STDP



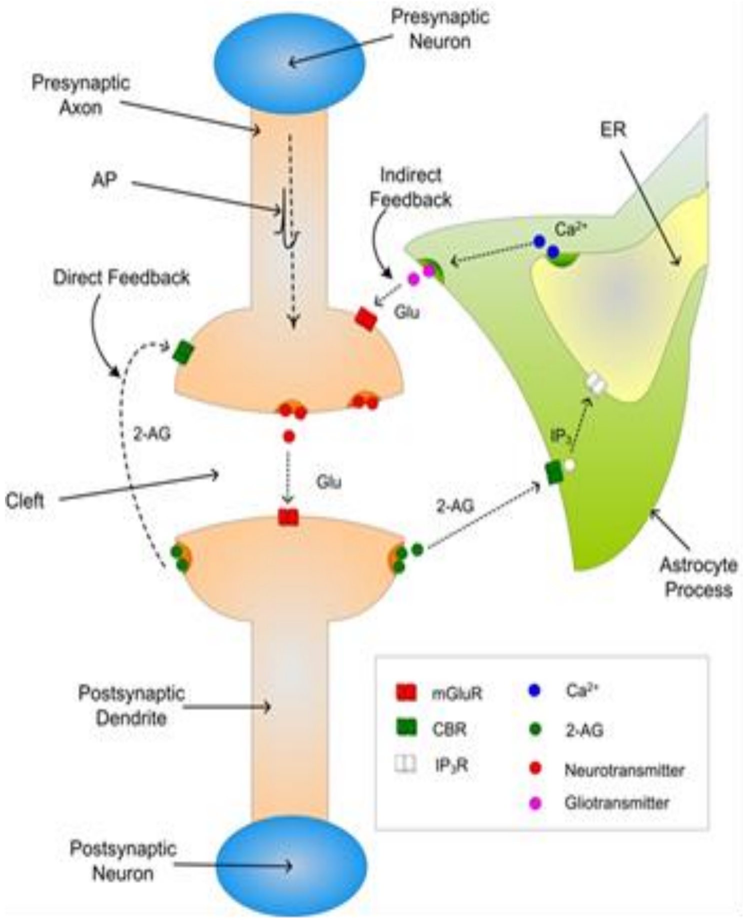

**Bidirectional Coupling between Astrocytes and Neurons Mediates Learning and Dynamic Coordination in the Brain: A Multiple Modeling Approach- 2011**

# GABA Switches from Low to High PR

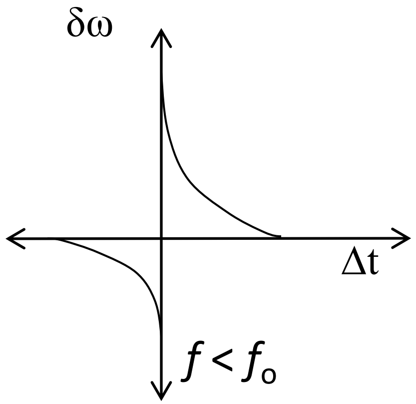


# BCM Rule Governs Plasticity

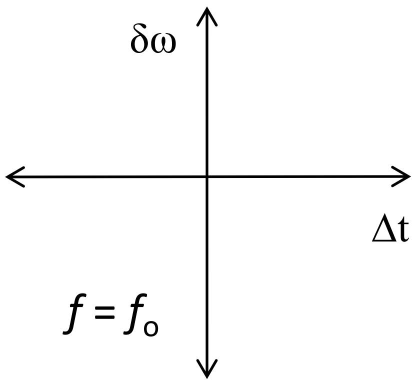



## Combined STDP-BCM rule – BSTDP rule




$$\delta\omega = A_o exp\left(\frac{\Delta t}{\tau_+}\right)\ for\ \Delta t \leq 0$$

$$\delta\omega = -A_o exp\left(\frac{\Delta t}{\tau_-}\right)\ for\ \Delta t > 0$$

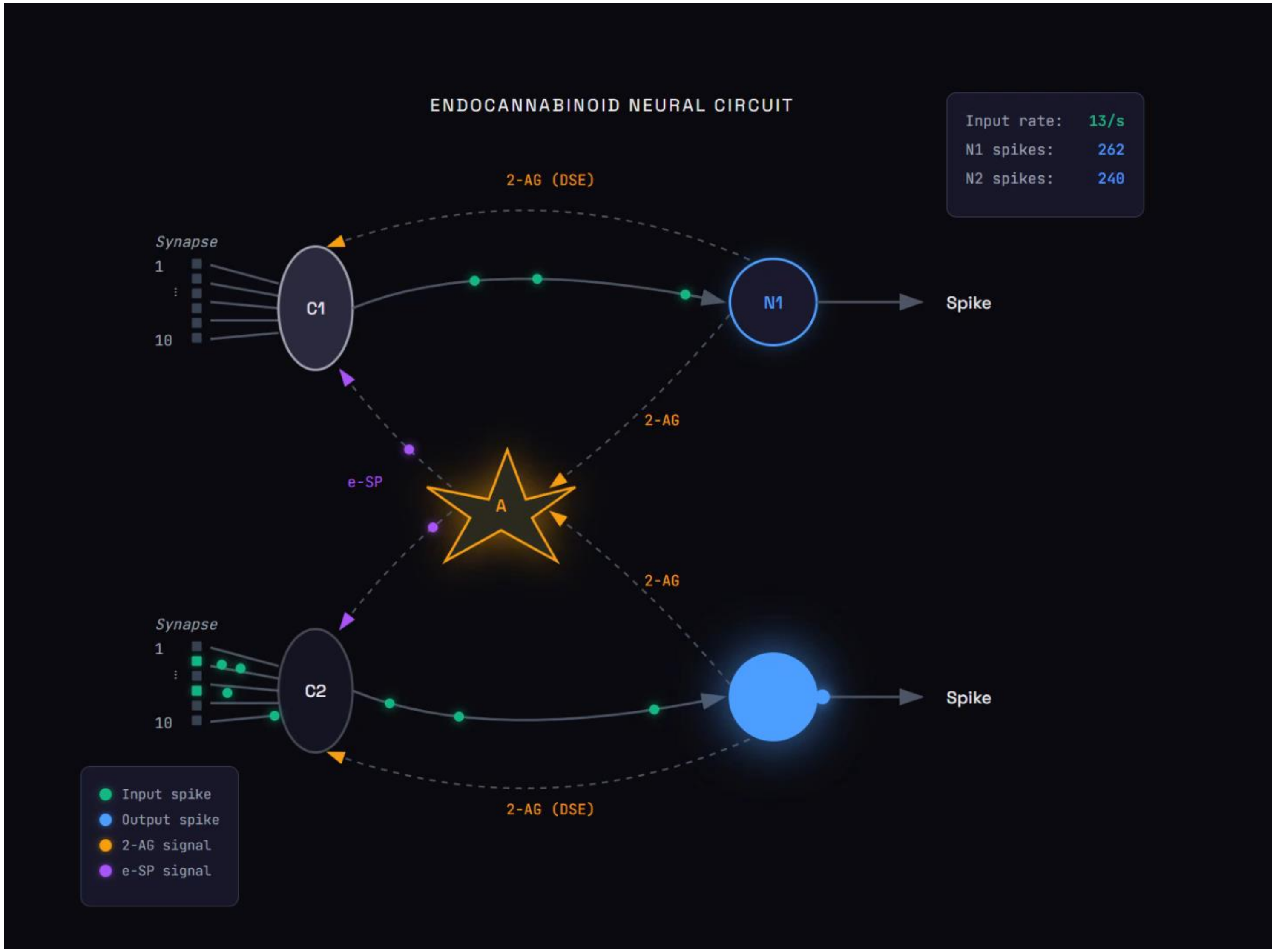



$$A_o = \frac{A}{1 + exp^{a(f-f_0)}} - A_-$$

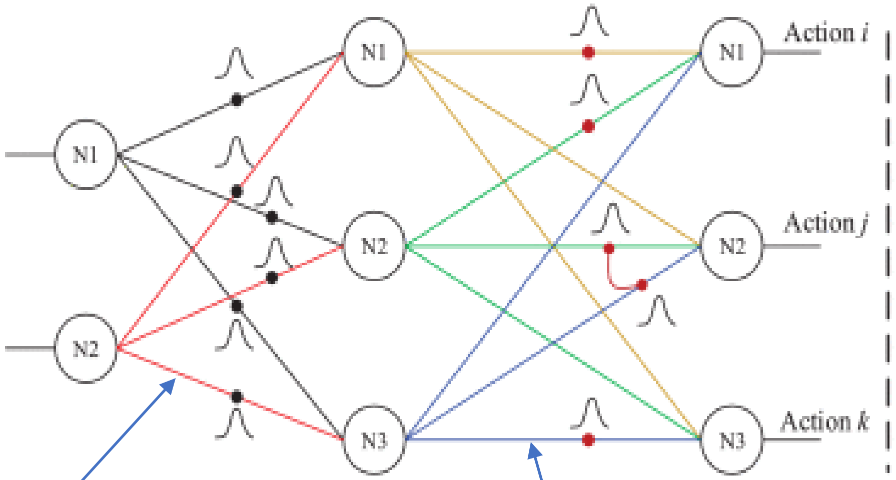
$f$= actual firing rate of postsynaptic neuron

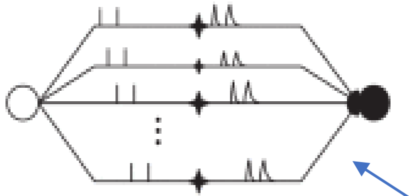
$f_o$ = target firing rate of postsynaptic neuron

# Endocannabinoid Pathway



# SANN Architecture with Multiple Pathways



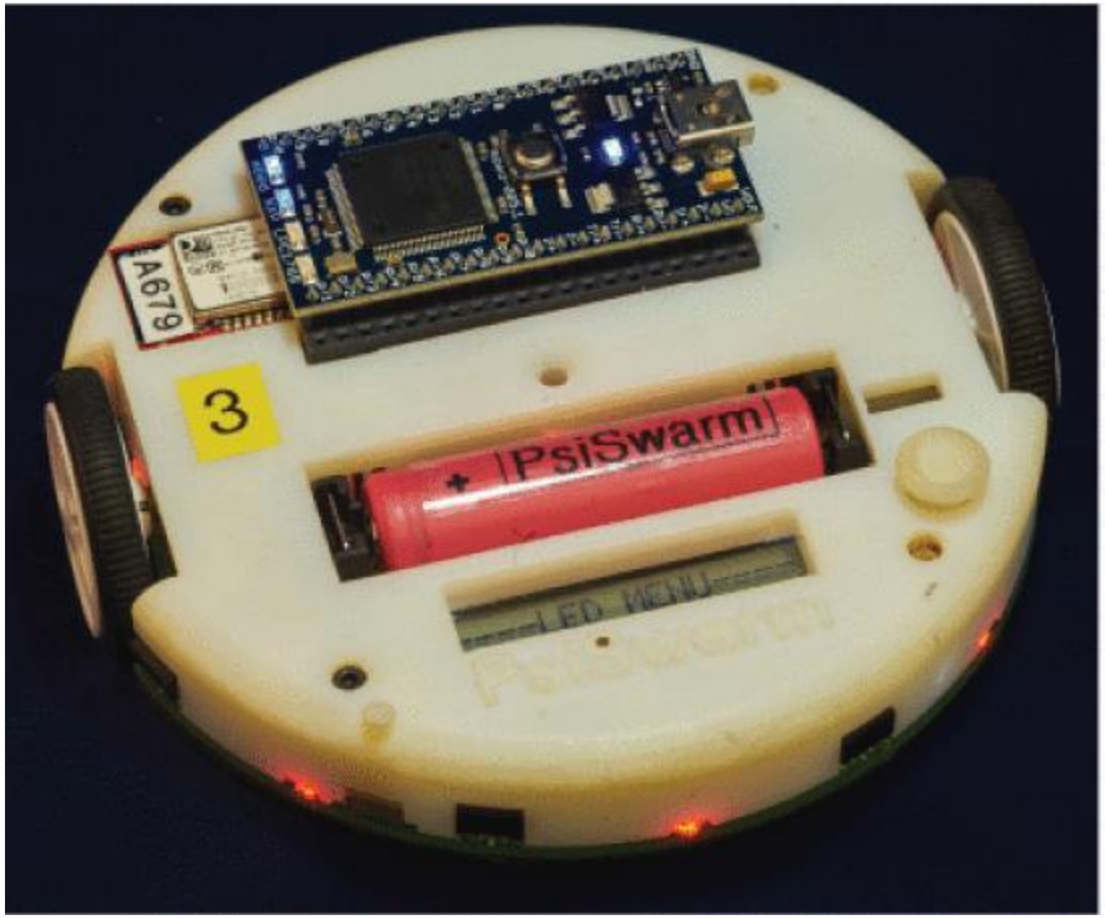
8 pathways - input to hidden layer

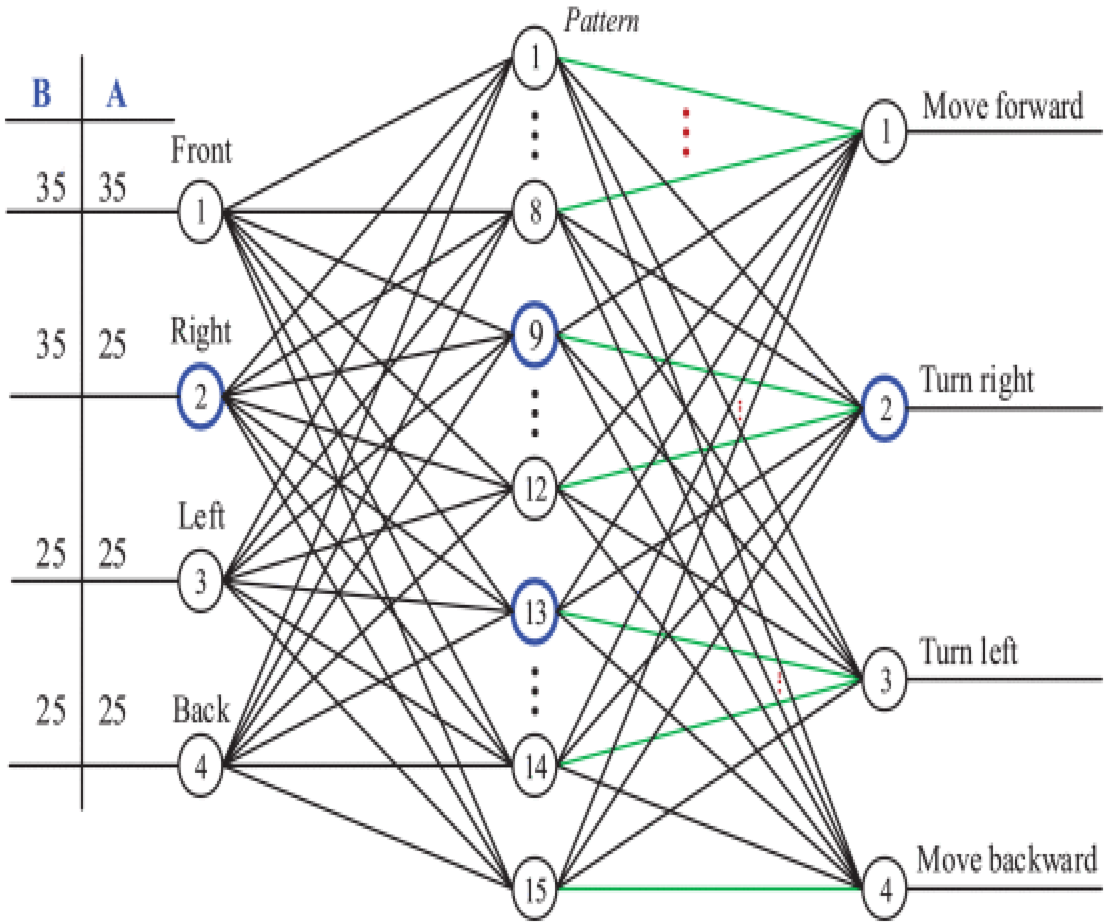
16 pathways - hidden to output neurons

Synaptic connection has multiple pathways

Each pathway is delayed by 1 ms from the neighboring pathway.

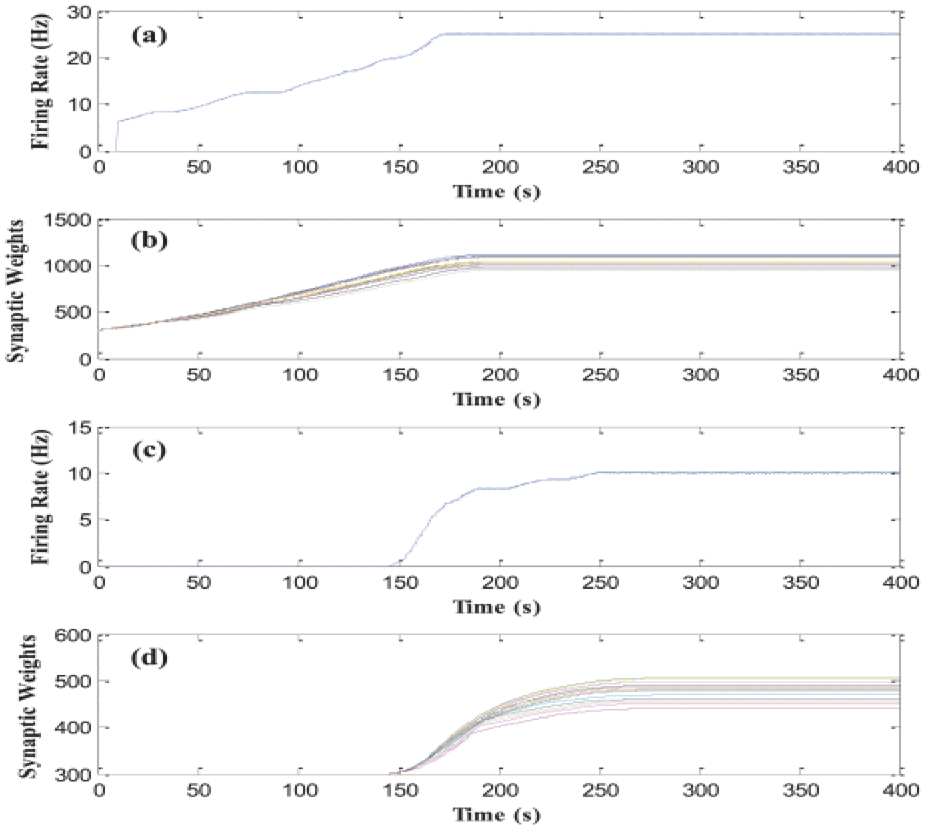
# PsiSwarm Robot with Controlling SANN



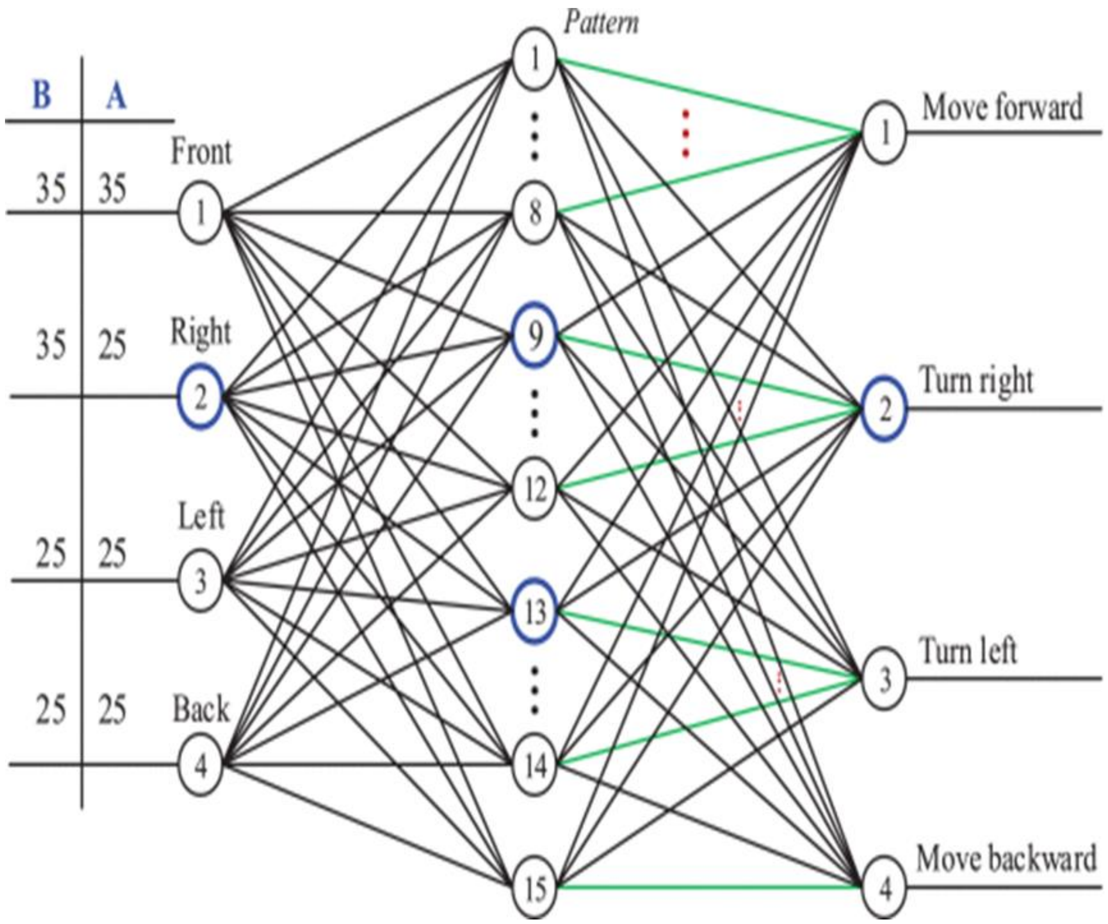

action priorities are F>R>L>B

# No Fault condition – Learning Phase



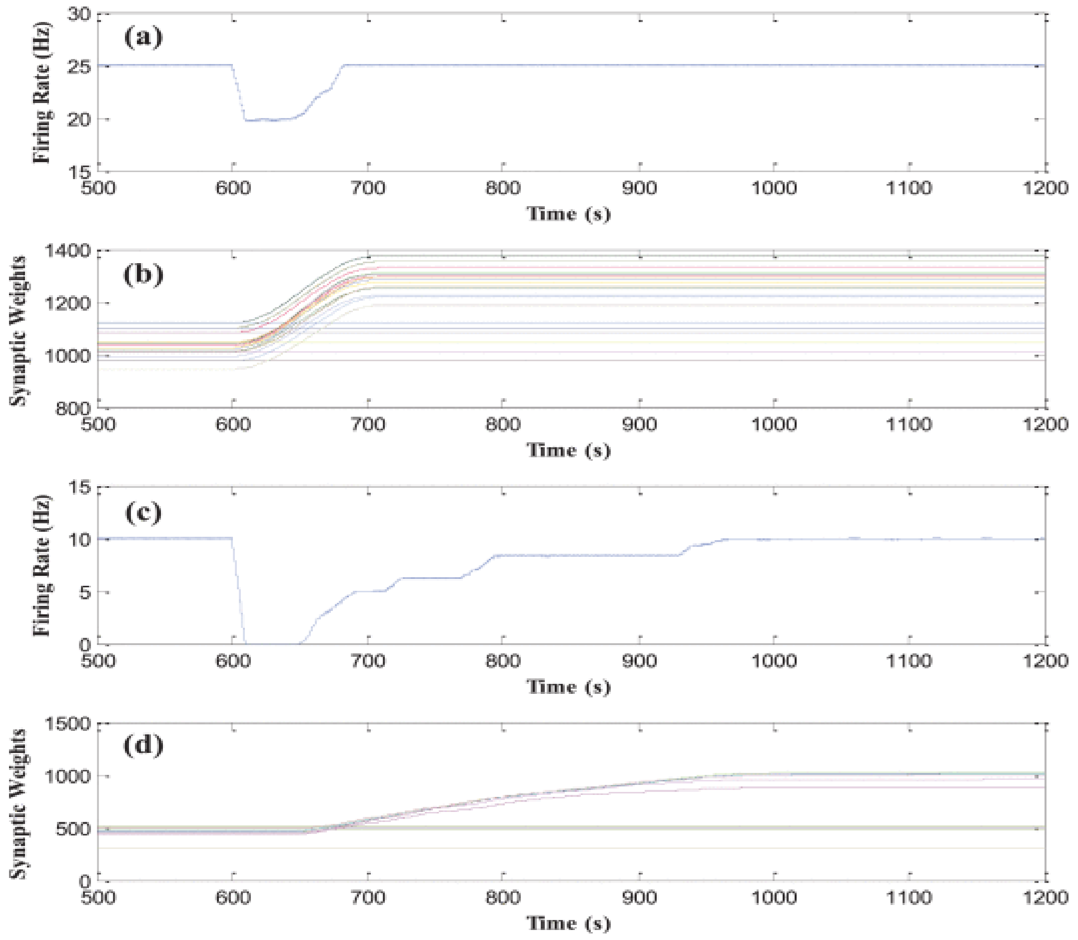

(a)/(b) - firing rate and synaptic weights of neuron #9 of the hidden layer

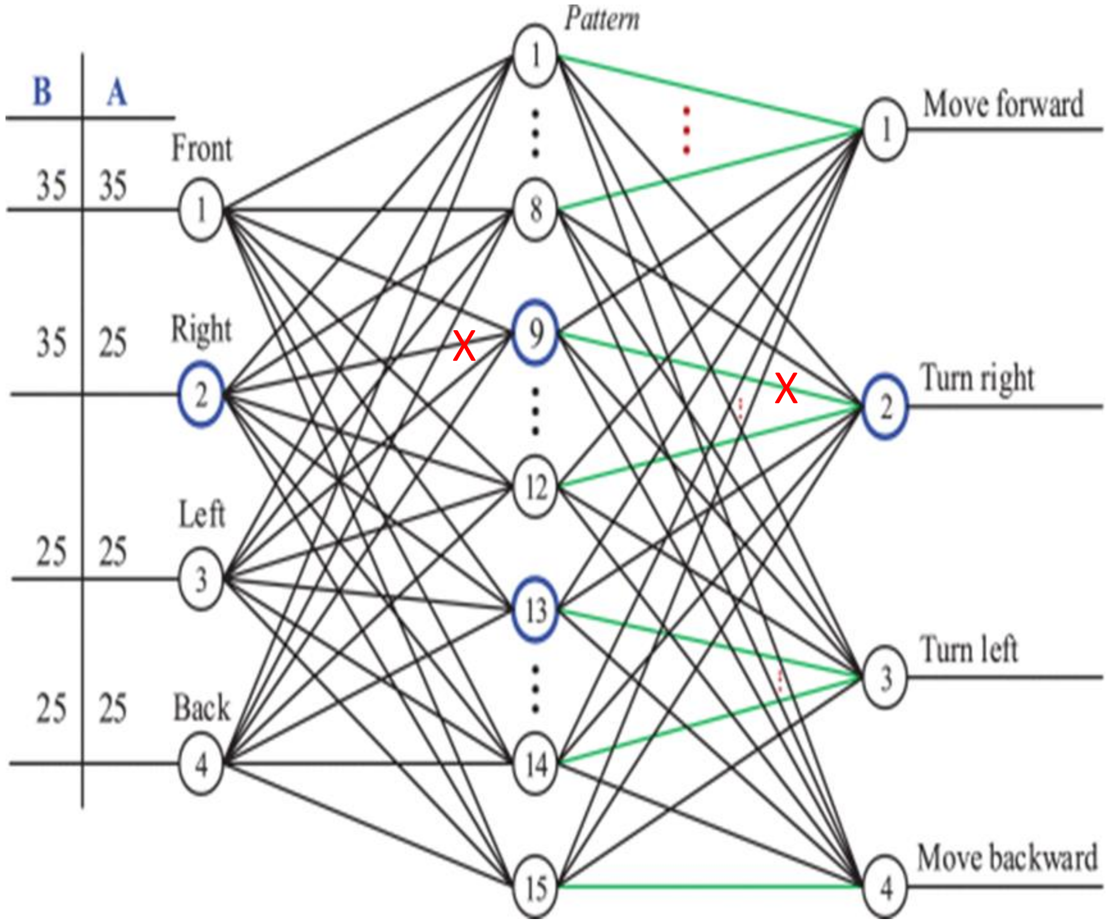
(c)/(d) - firing rate and synaptic weights of neuron #2 of the output layer.




Obstacle in front of the robot – Turn Right

# Fault Condition – 50% in Both Layers

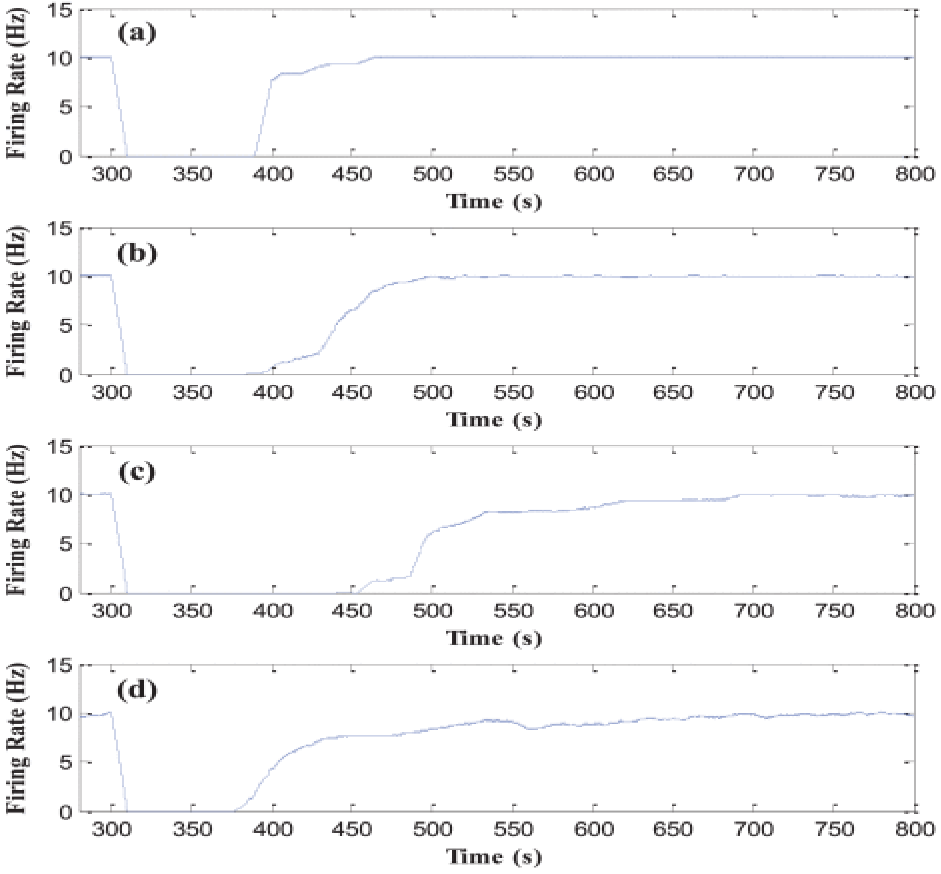


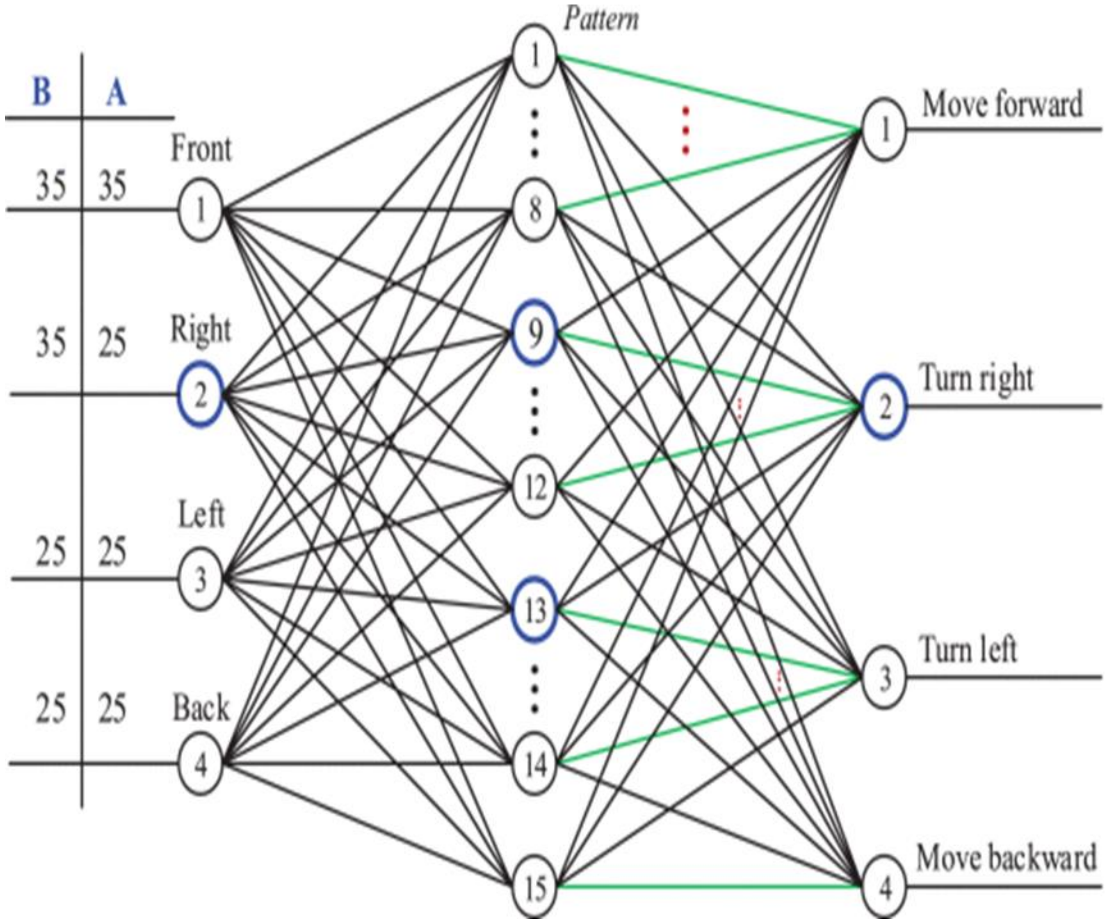




(a)/(b) - firing rate and synaptic weights of neuron #9
(c)/(d) - firing rate and synaptic weights of neuron #2

X – 50% of the pathways develop faults – PR=0

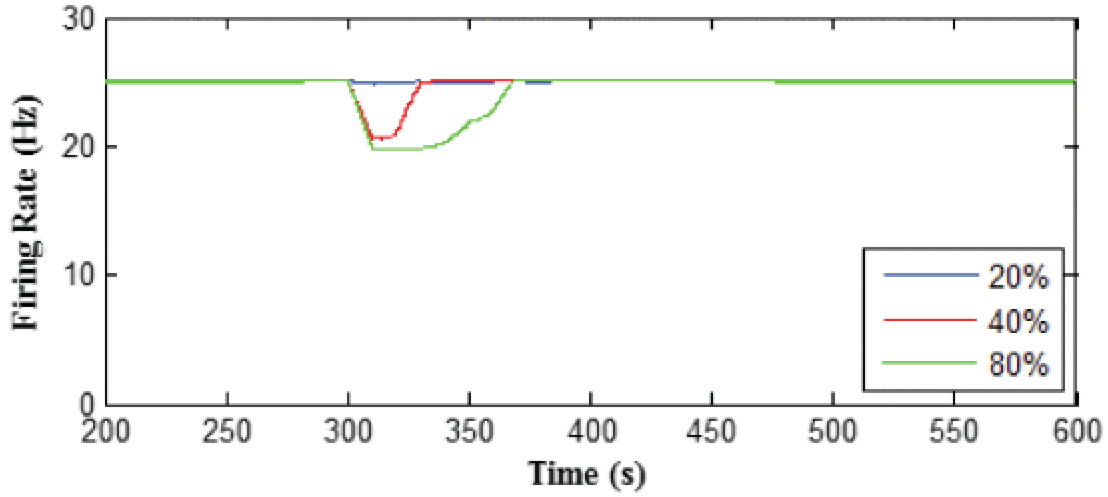
# Random Fault Distribution – 40%



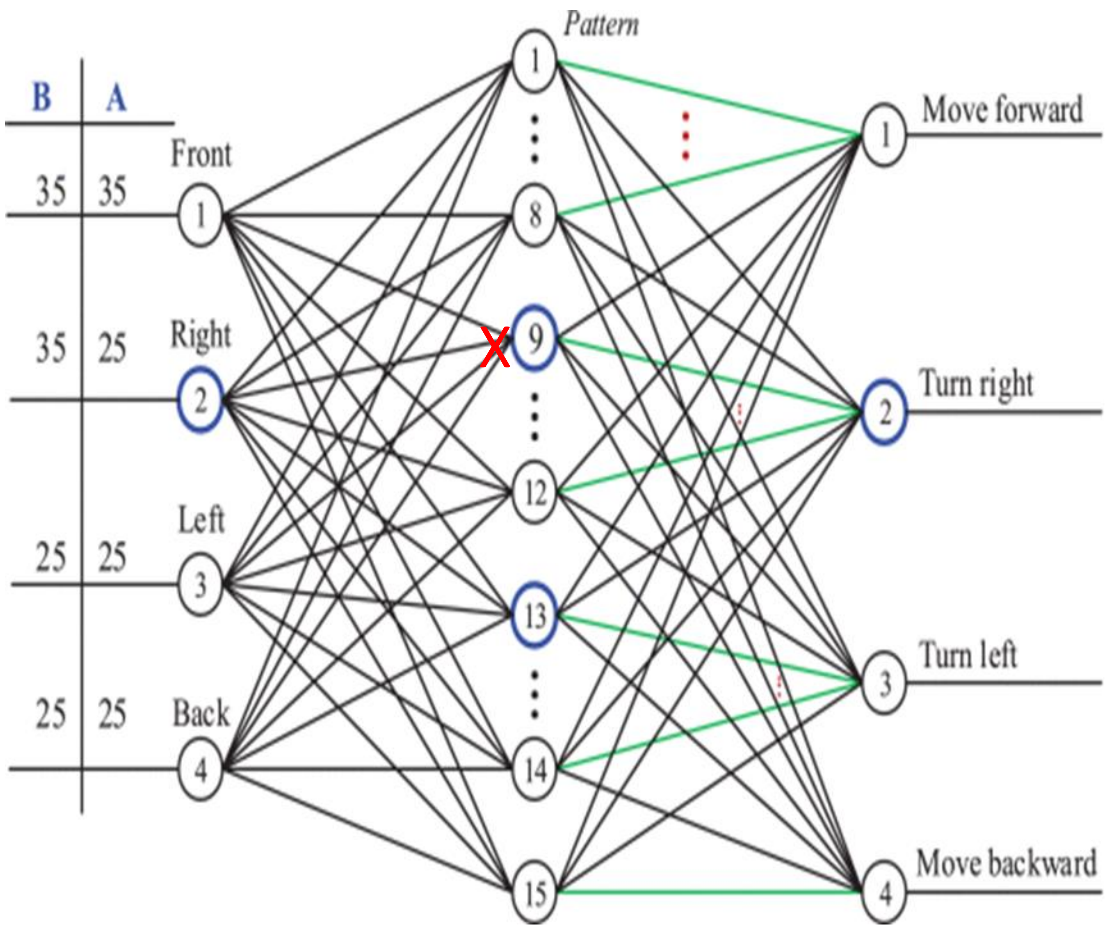




(a)/(d) - firing rates of neuron #1–#4 of the output layer

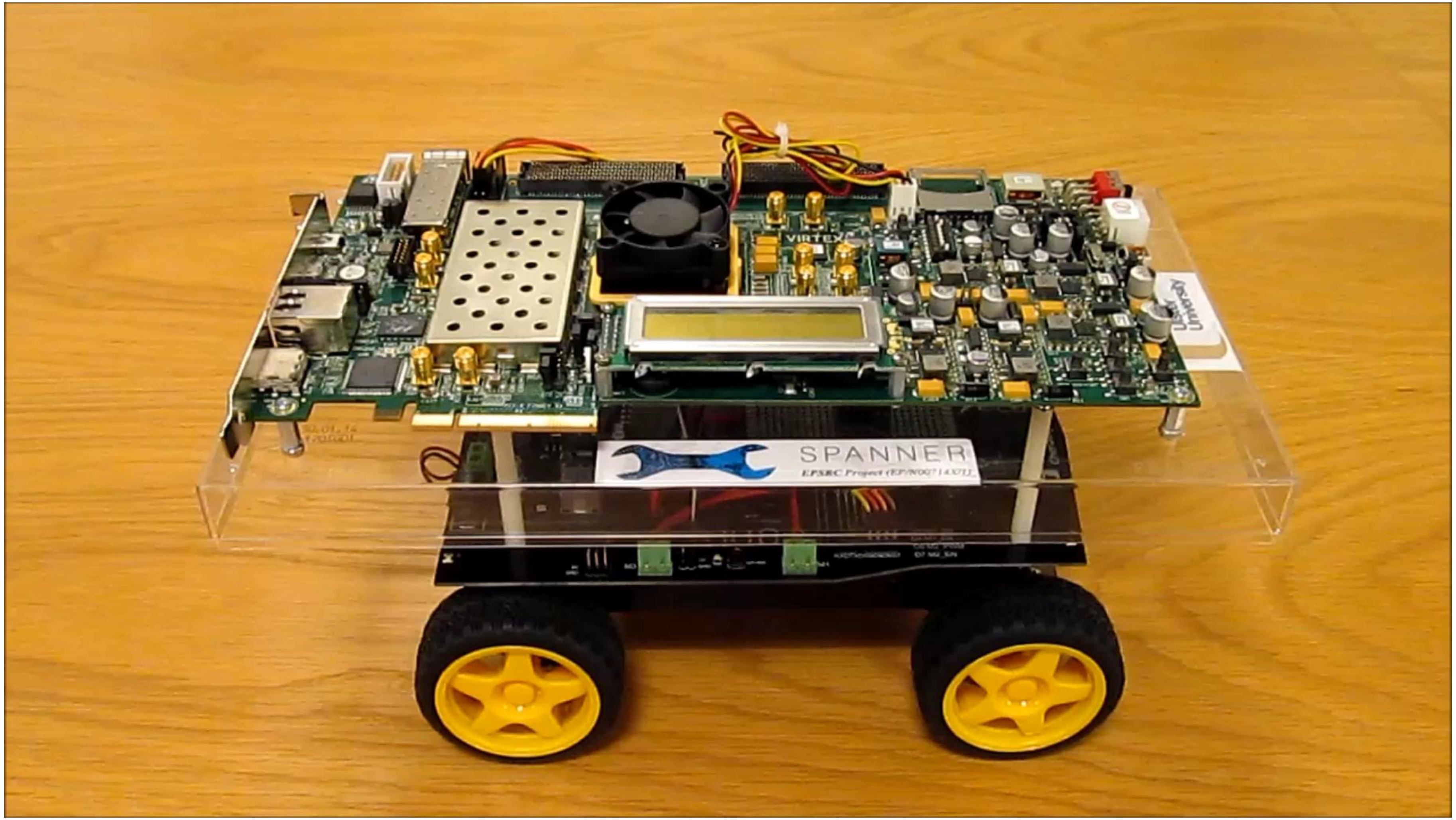
# Fault Condition – 20% to 80%



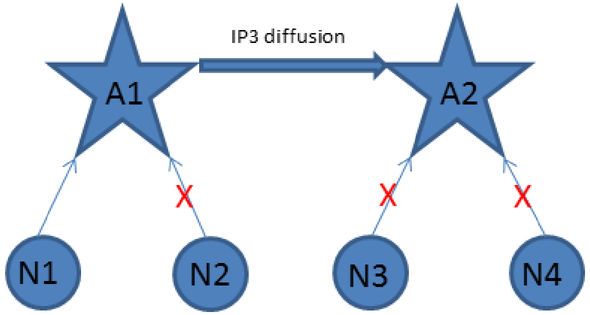

Firing rate of the neuron #9 under various fault densities

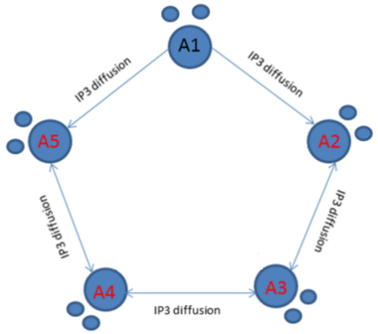


# Self Repairing Hardware Demo



# Distributed Self Repair



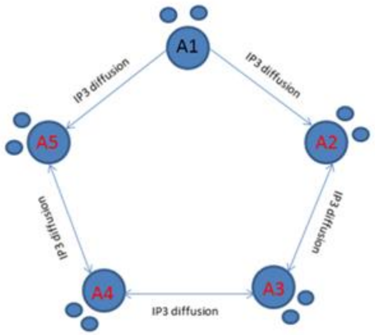



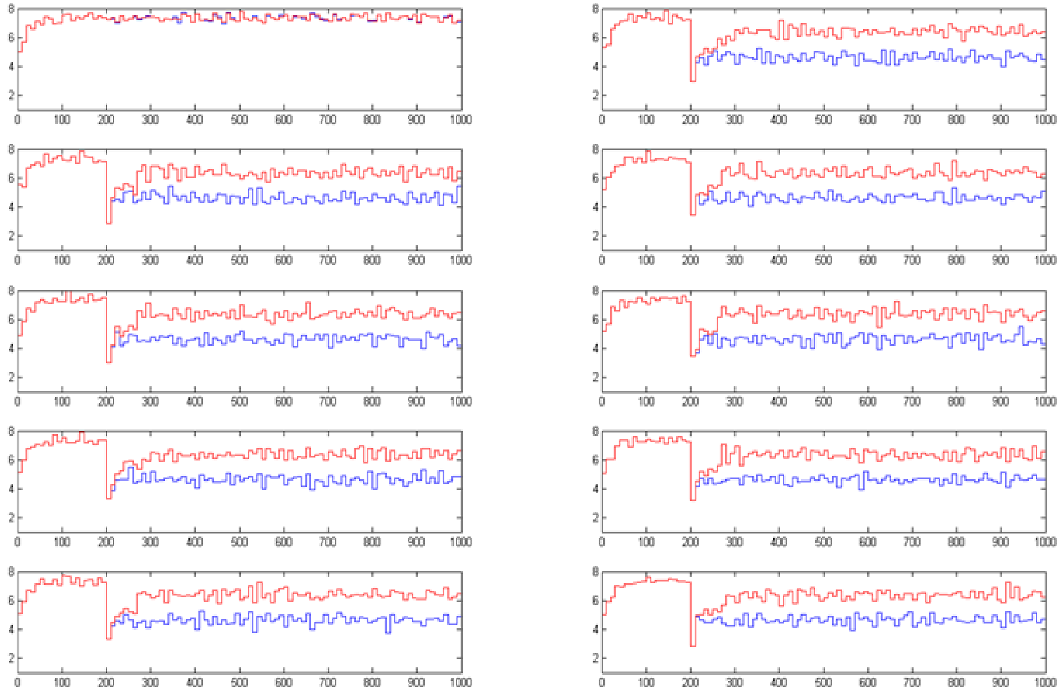

$$J_{i \to j} = F \cdot \Delta_{ij} IP_3$$

$$\Delta_{ij} IP_3 = IP_3^{\,i} - IP_3^{\,j}$$

M. Goldberg, M. De Pittà, V. Volman, H. Berry and E. Ben-Jacob, “Nonlinear Gap Junctions Enable Long-Distance Propagation of Pulsating Calcium Waves in Astrocyte Networks”, PloS Comput. Biology, Aug 26, 2010

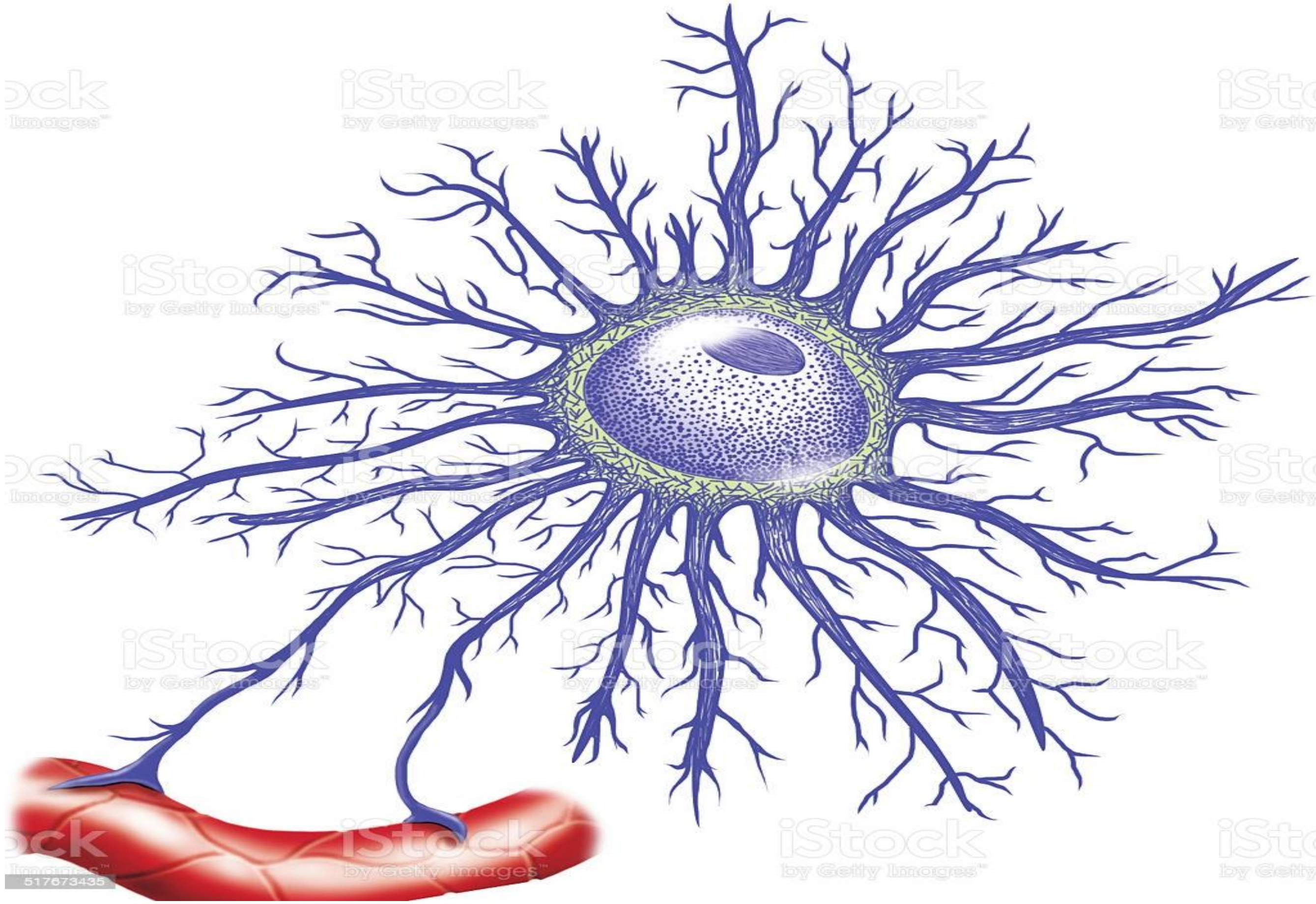
# Distributed Self Repair



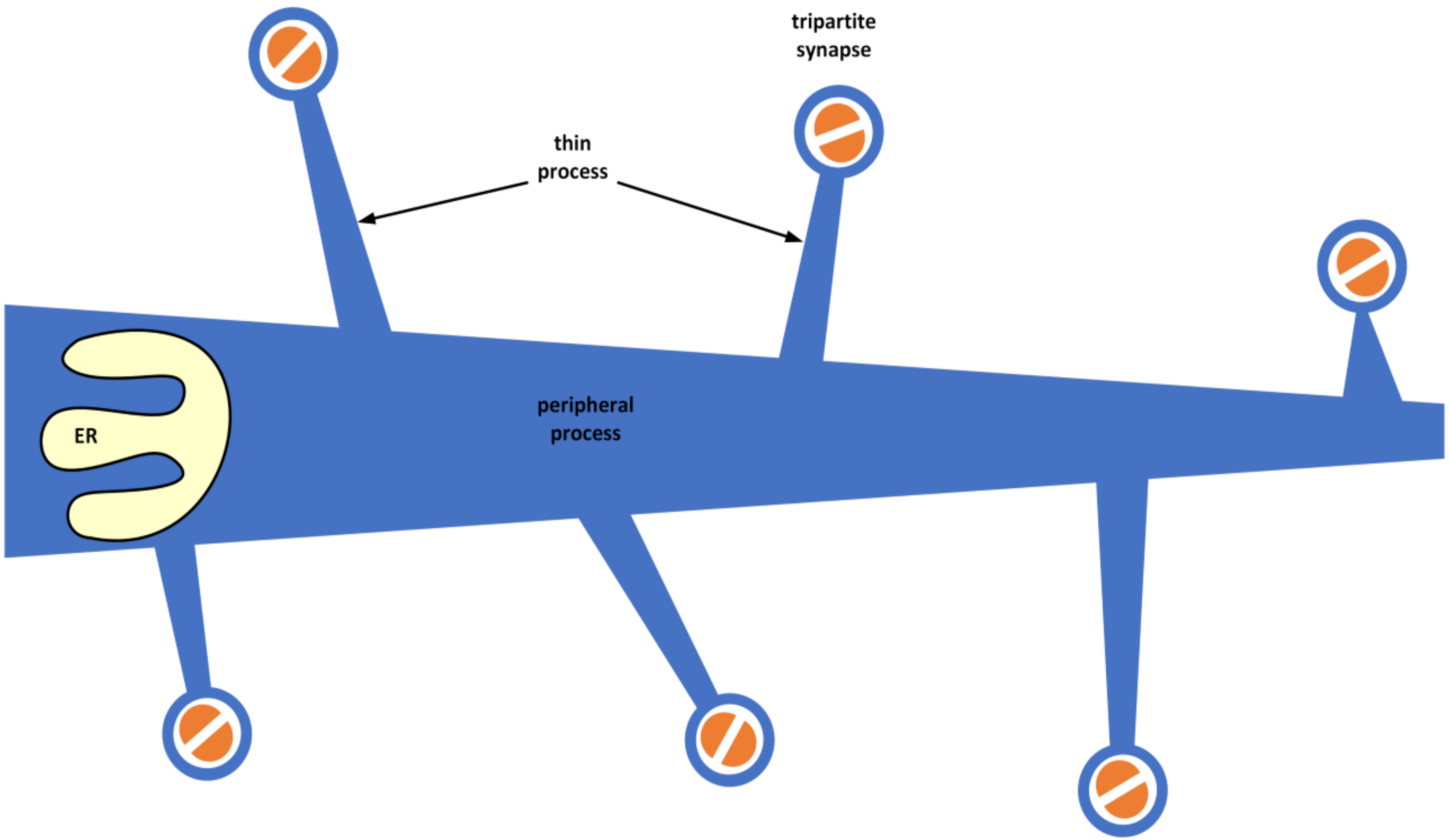



**Firing rates of 5-Astrocytes in a ring network with BCM-STDP module (Red) and without (Blue).** Row 1: A1 (N1, N2) - N1 is healthy and N2 is 80 % damaged. Row 2: A2 (N3, N4) - both 80 % damaged. Row 3: A3 (N5, N6) - both 80 % damaged. Row 4: A4 (N7, N8) - both 80 % damaged. Row 5: A5 (N9, N10) - both 80 % damaged

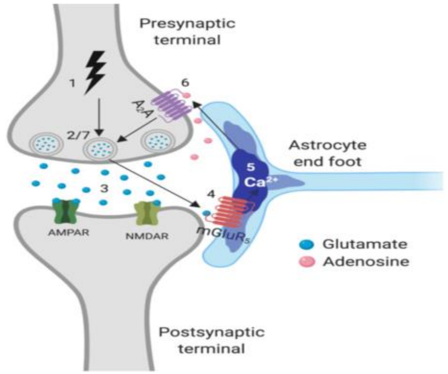
# Astrocyte Primary & Thin Processes

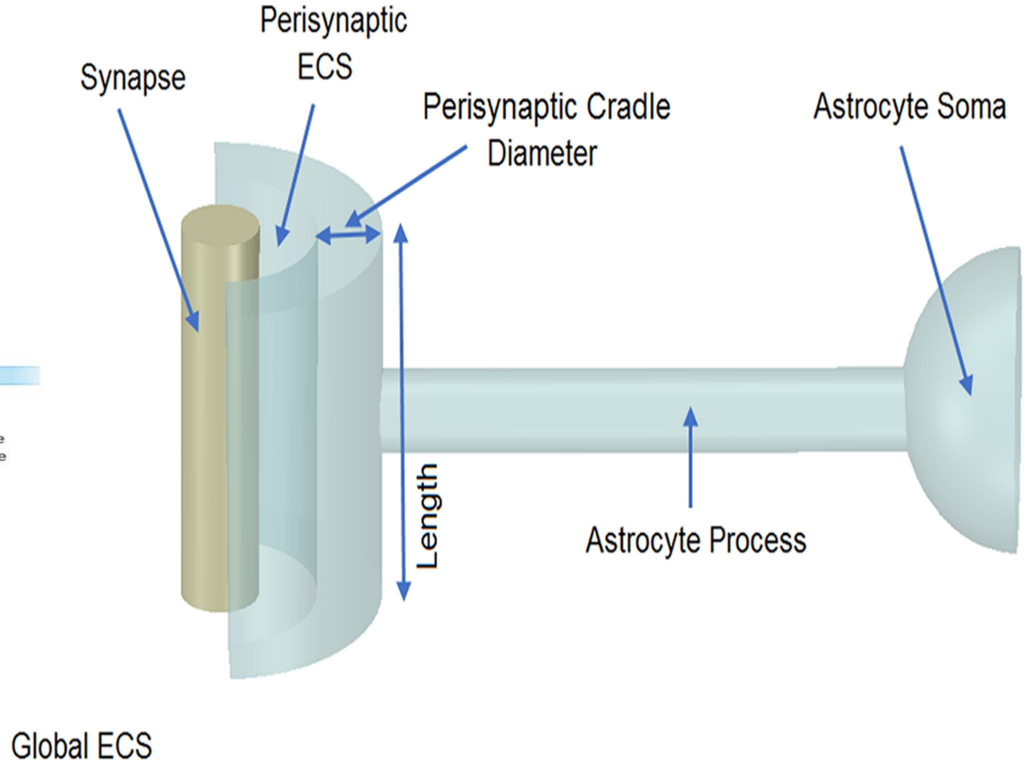


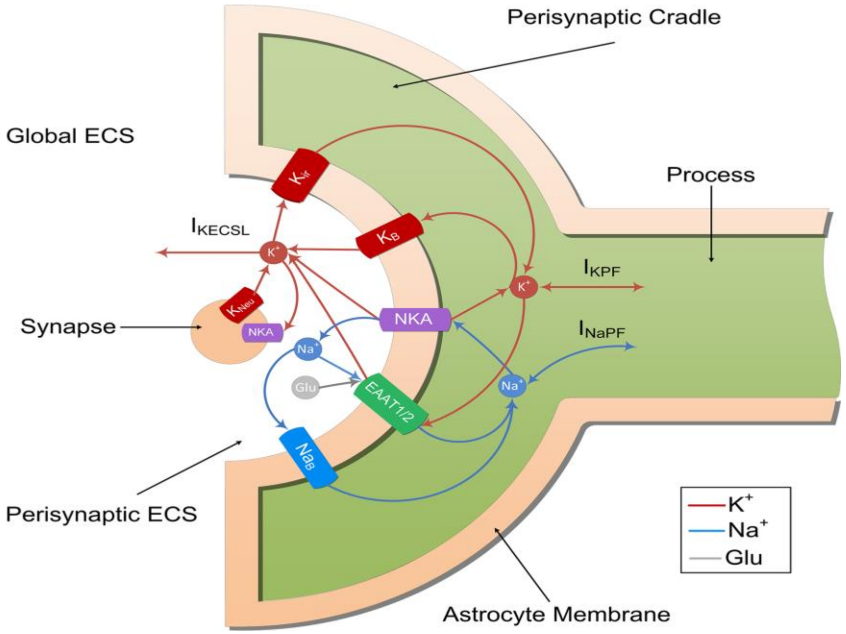
# Model of Primary & Thin Processes



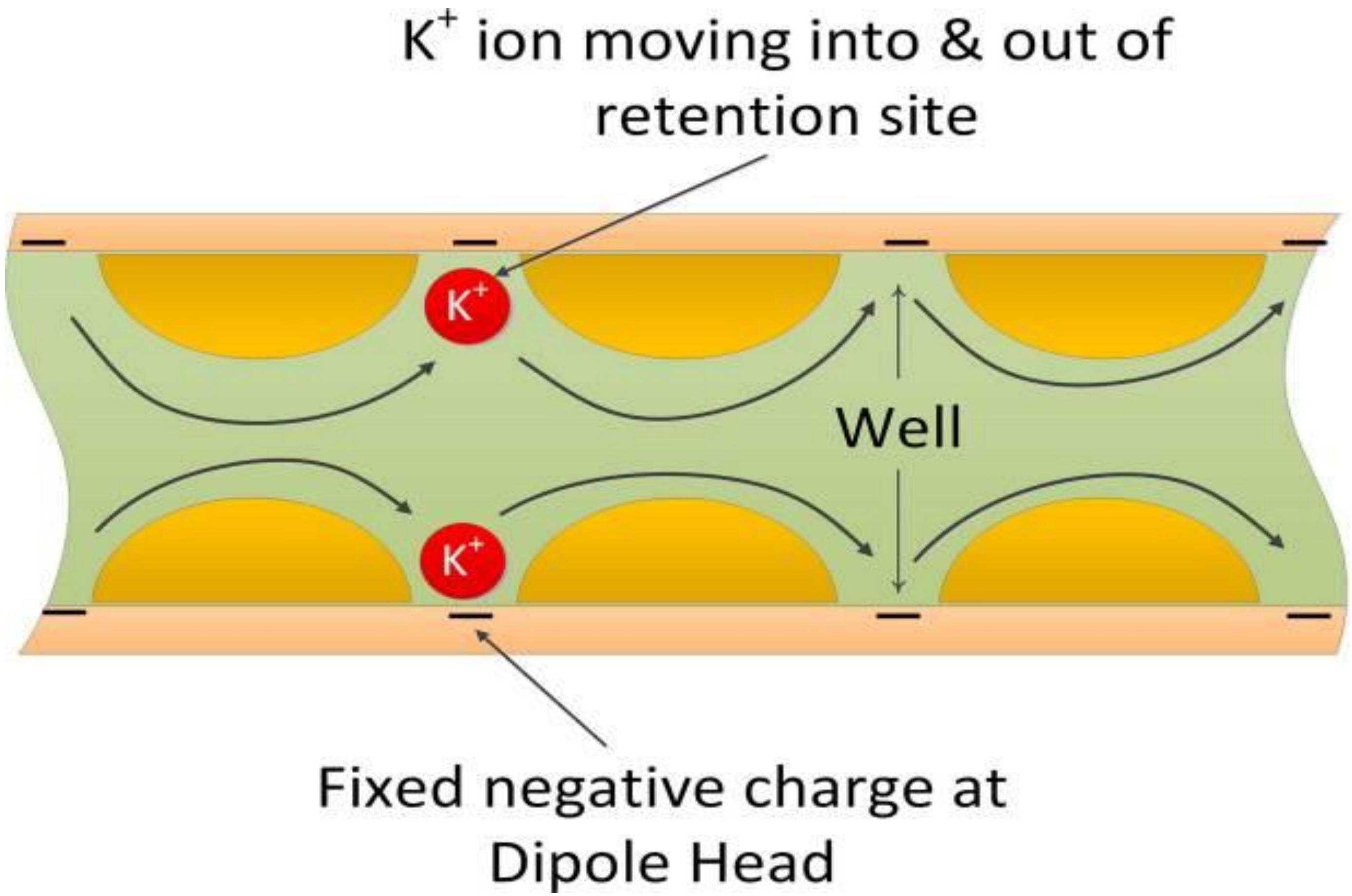
# Tripartite Synapse Model



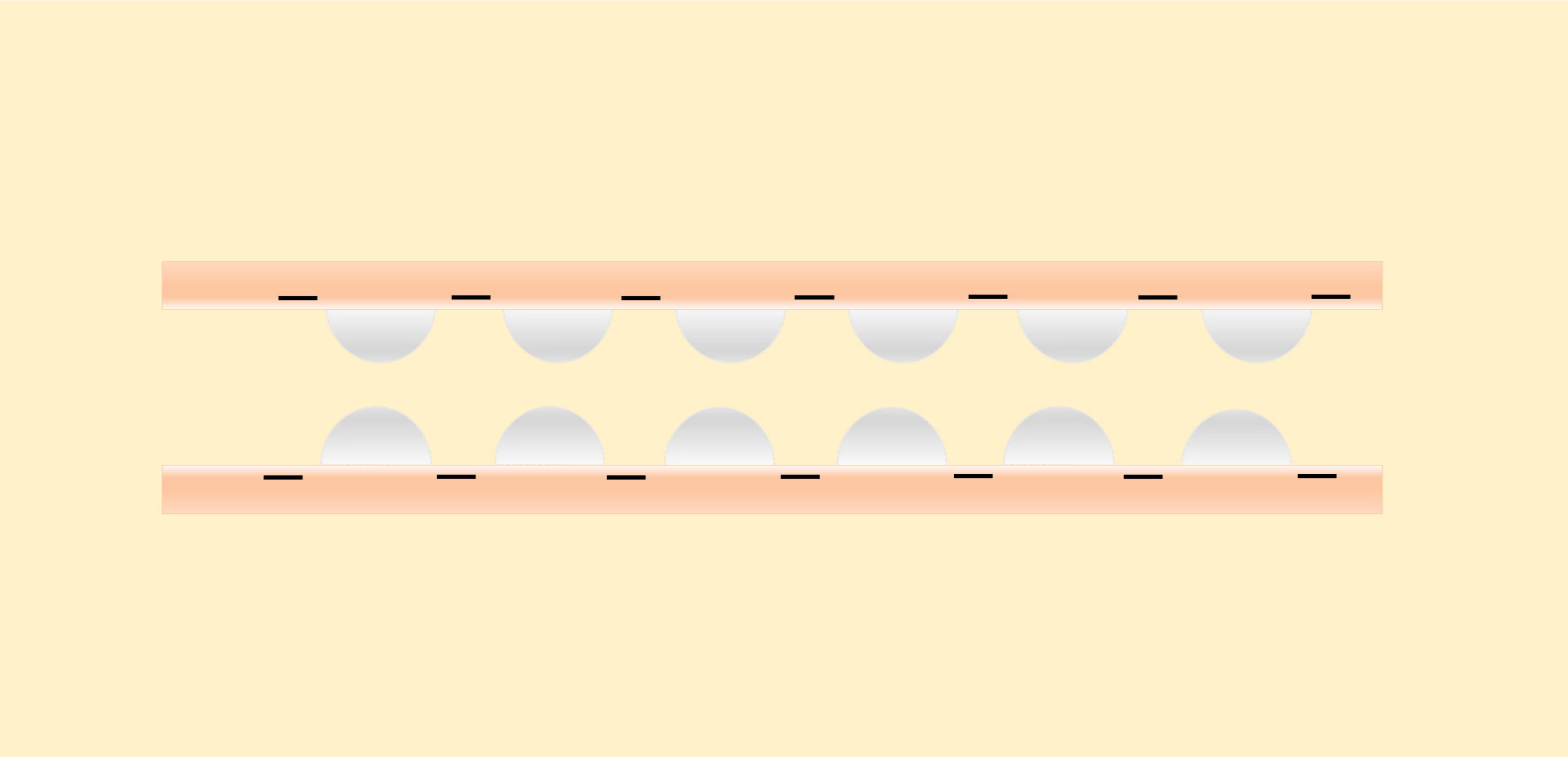



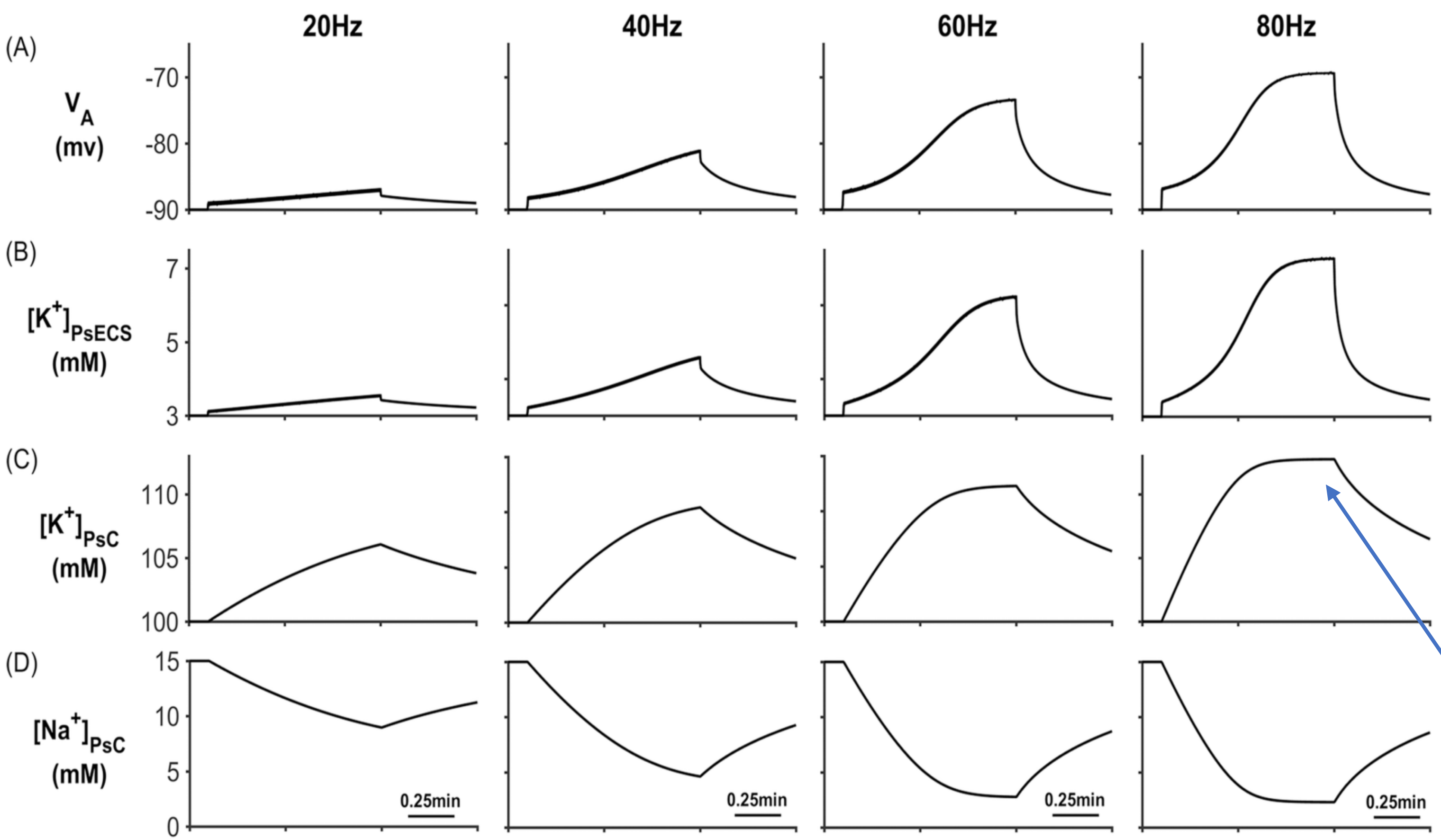
# Retention Sites in Thin Processes

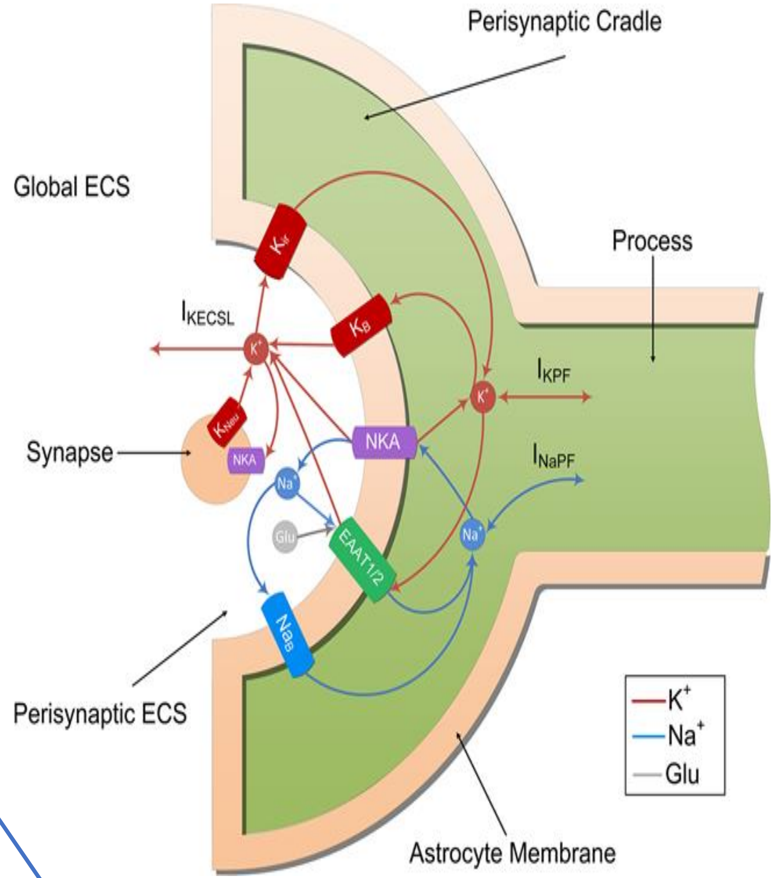


# Retention Sites in Thin Processes



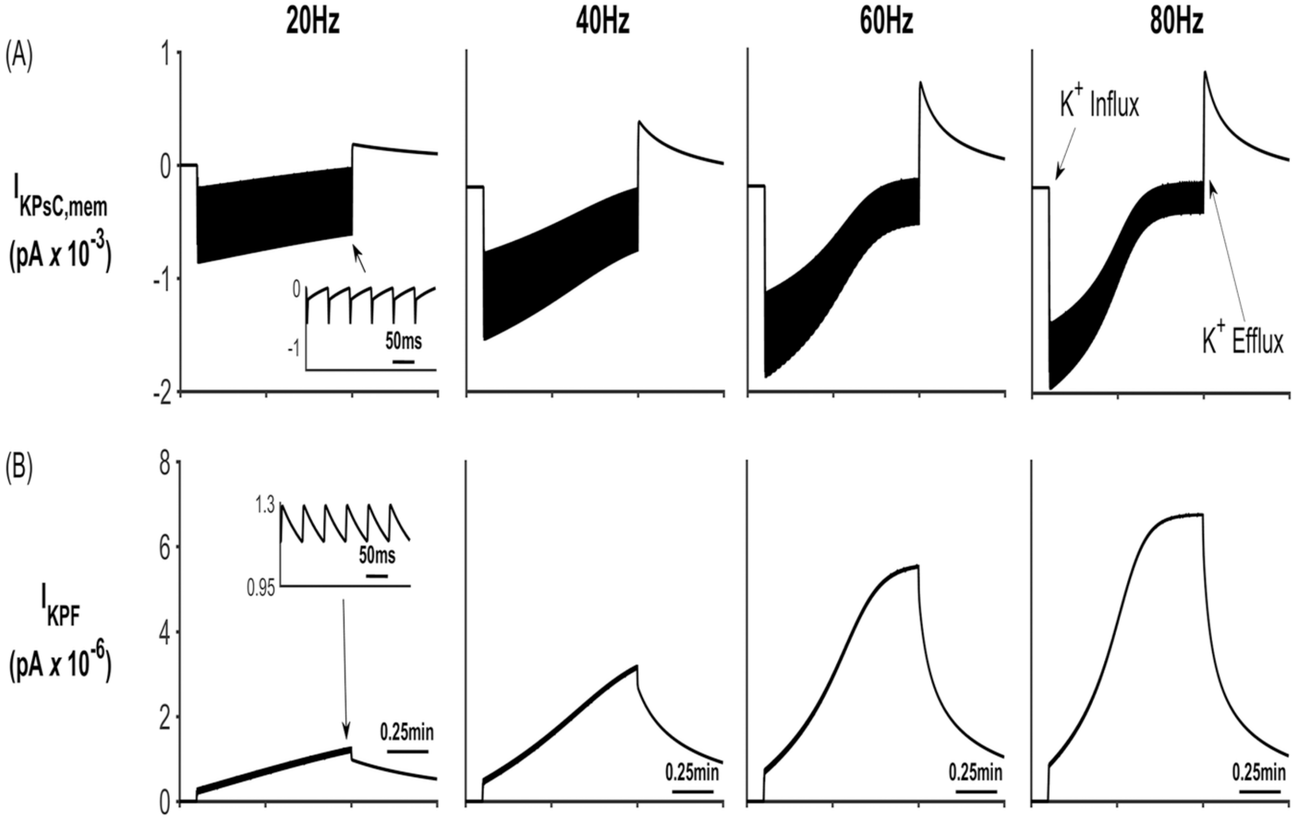
# Potassium Buffering

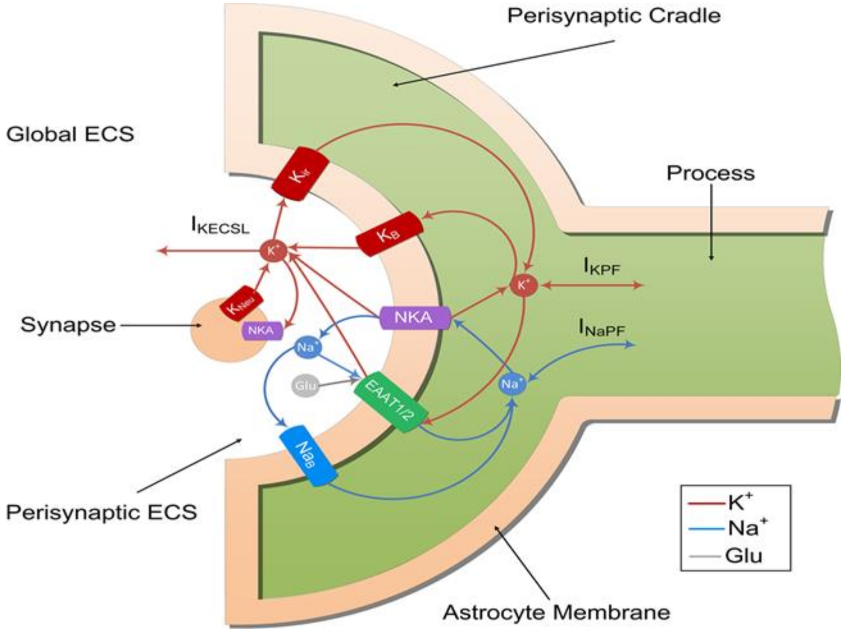



Microdomain

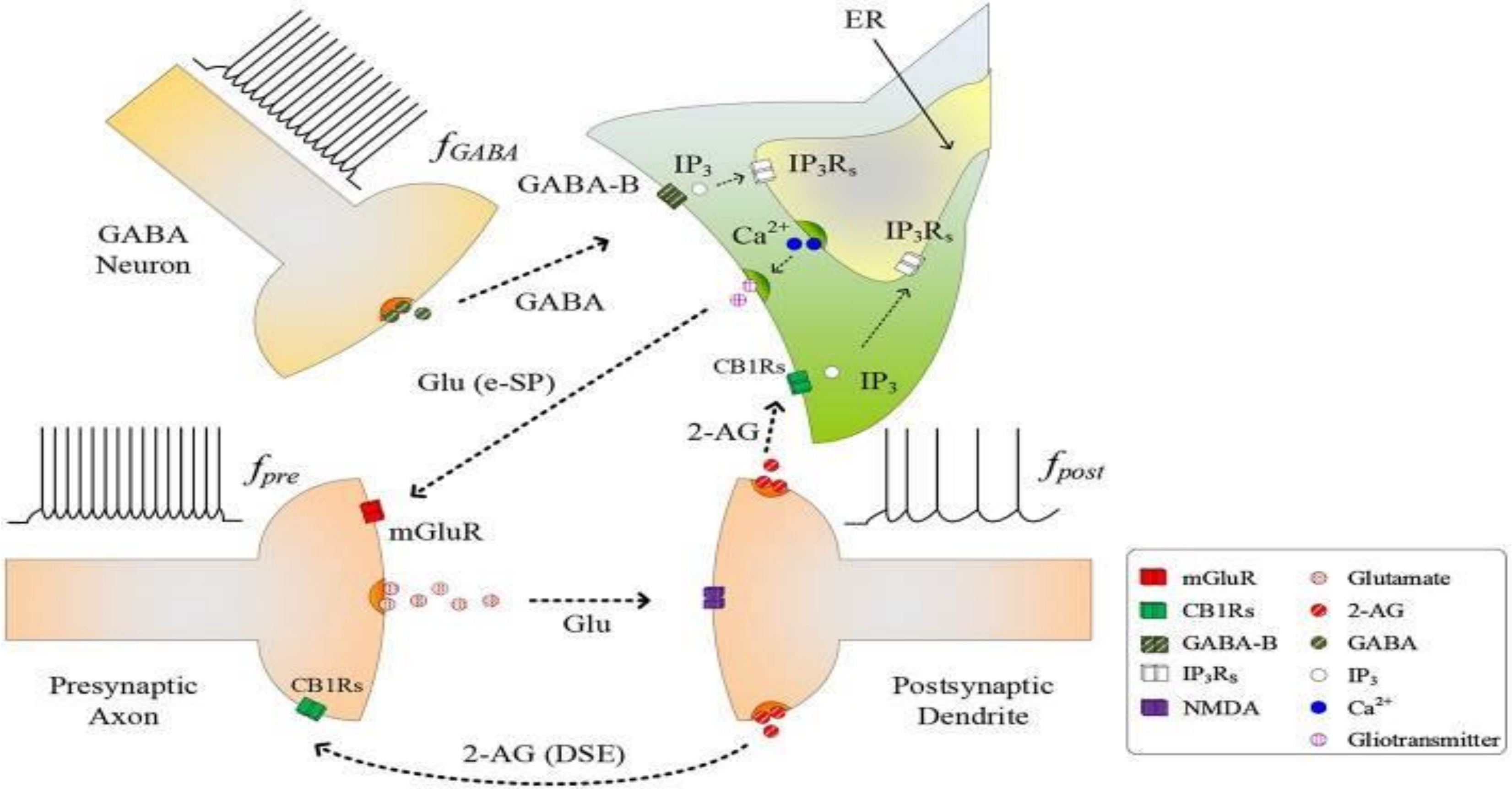
# Potassium Buffering



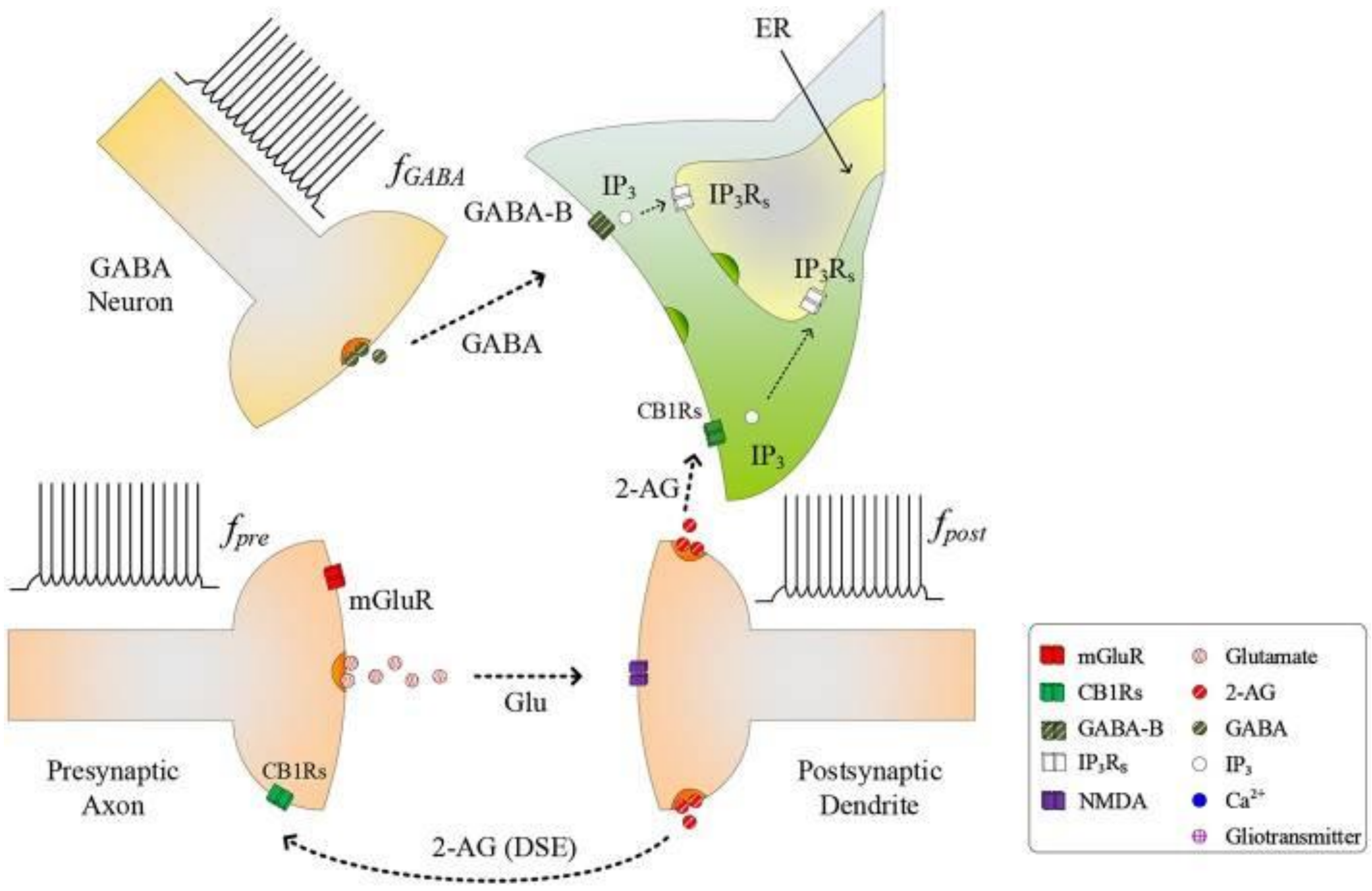




Potassium and sodium microdomains in thin astroglial processes: A computational model study - 2018

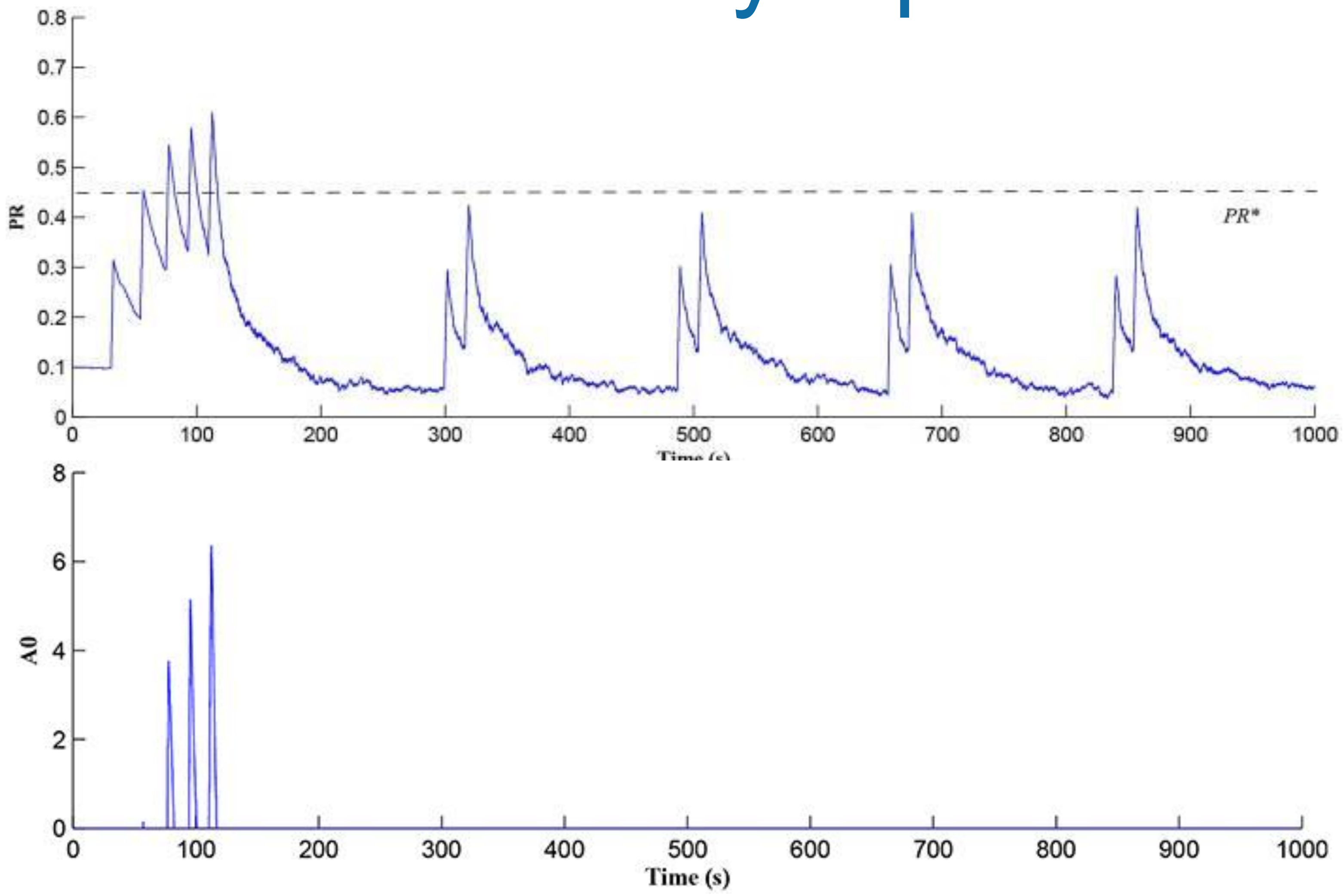
# GABA Interneurons - Tripartite synapse

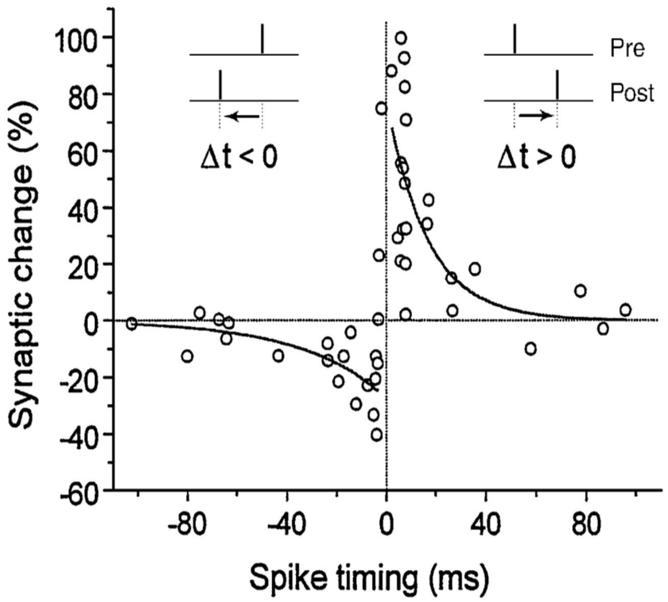


# GABA Interneurons - Tripartite synapse

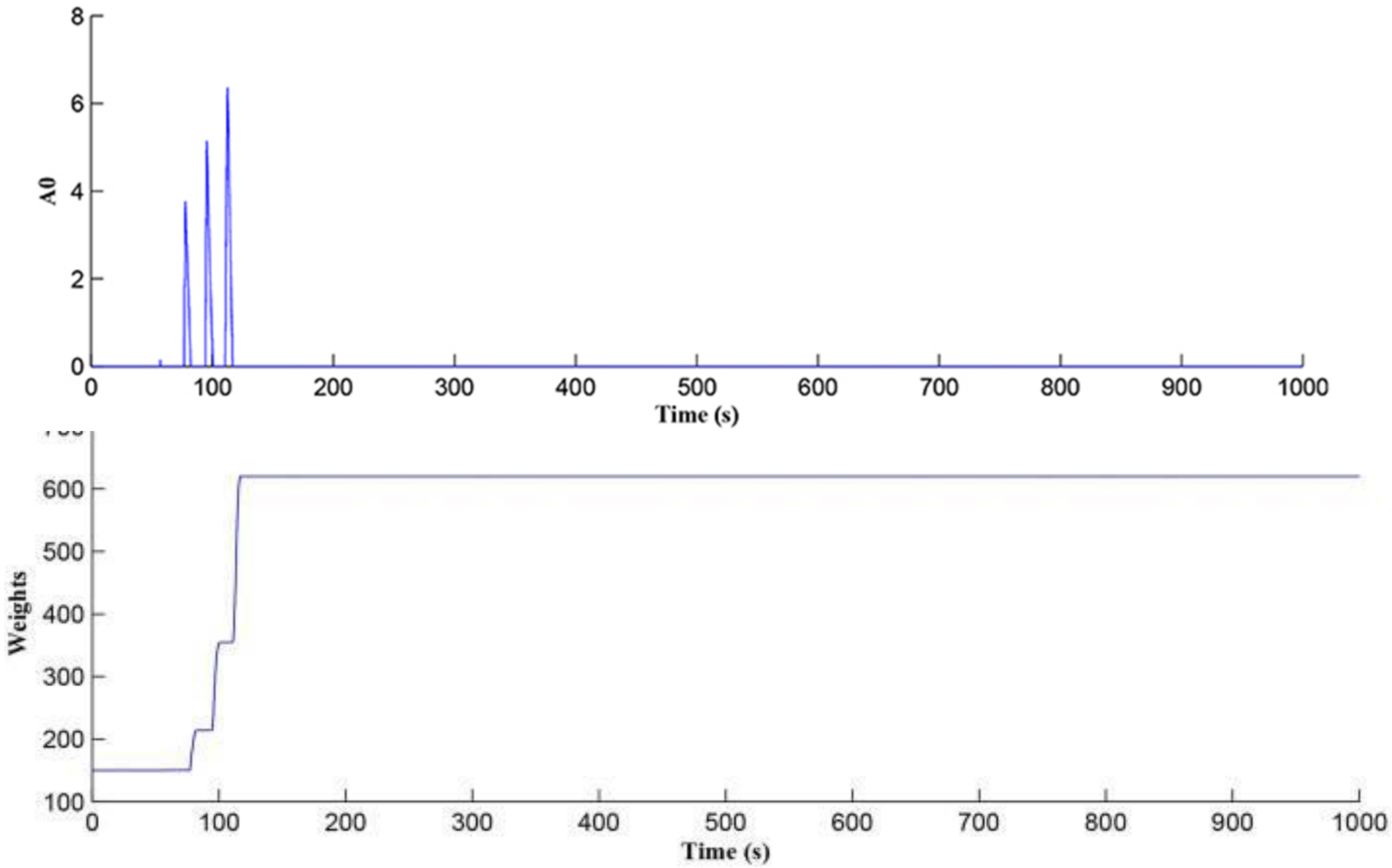


# STDP - Synaptic Potentiation

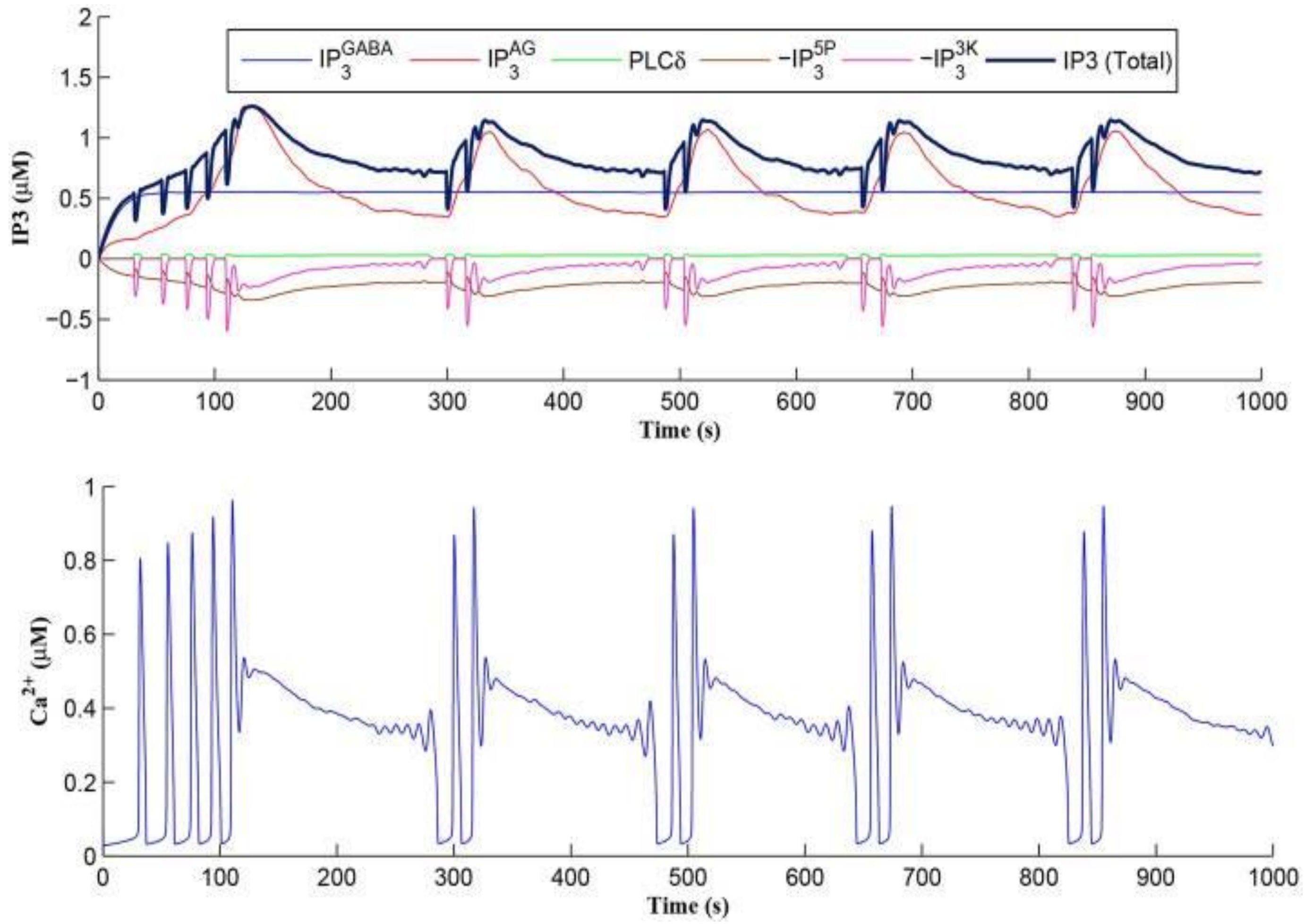


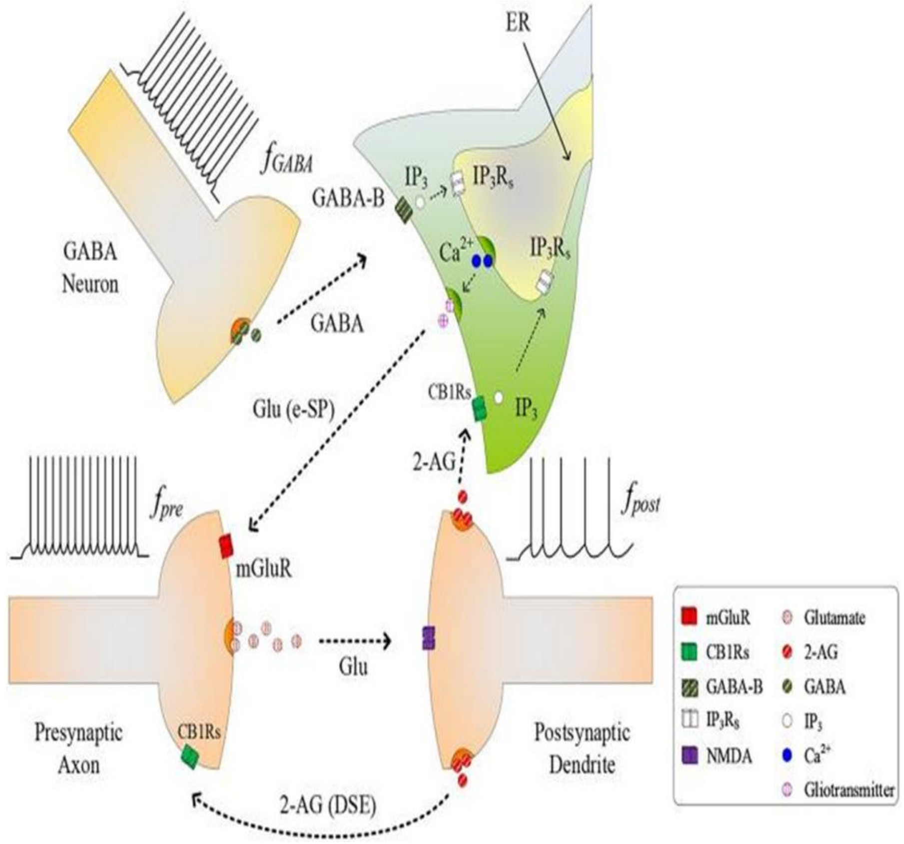



# Weight Potentiation When STDP Window Open

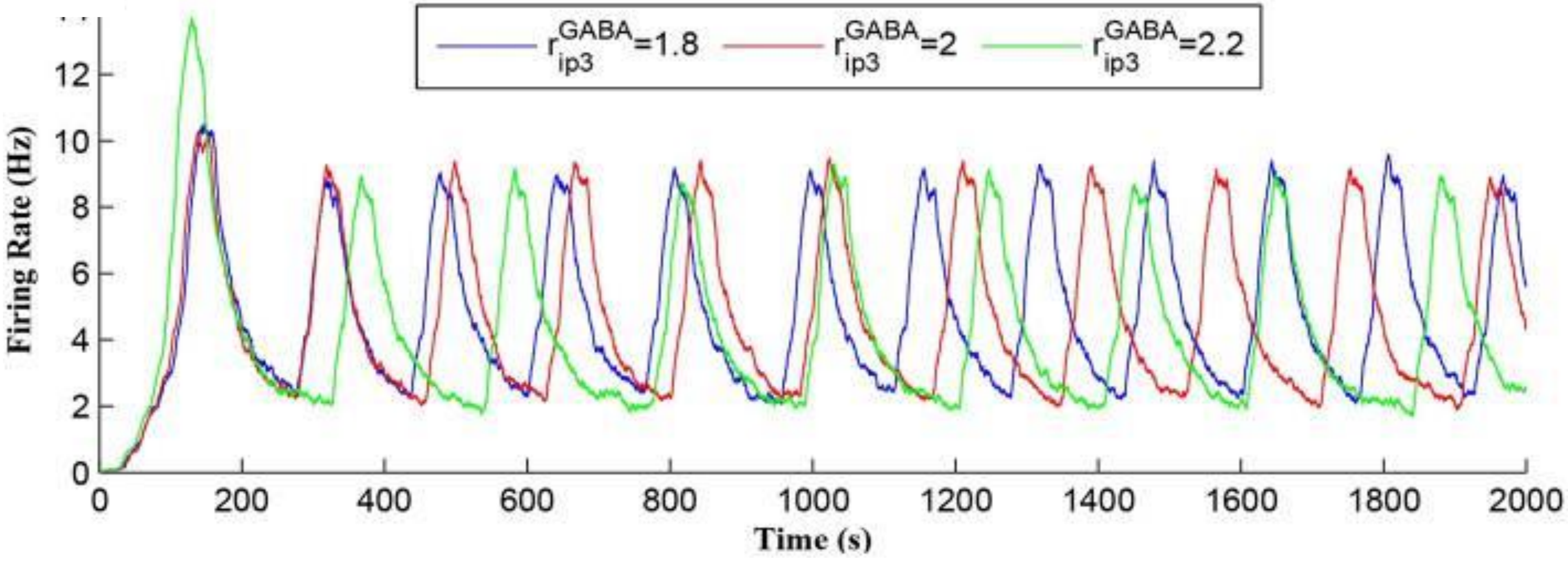


# Calcium Oscillations



# Neuronal Burst Firing




Change in GABA production rate $r_{ip3}^{GABA}$ leads to a change in burst firing rate –look at the first 1000 seconds - $r_{ip3}^{GABA} = 1.8$ gives 6 bursts, $r_{ip3}^{GABA} = 2$ gives 5 bursts while $r_{ip3}^{GABA} = 2.2$ gives 4 bursts

# Conclusion

- Astrocytes - mainstream research in functional & dysfunctional brain

- Astrocytes – point to a novel NeuroAI research focus:

  - Ionic homeostasis at the tripartite synapse
  - Repair at cellular level – combined STDP/BCM rule
  -
  - Potassium Clearance & Cradle Storage
  - Dynamic co-ordination of neurons - oscillations

# Computational Neuroscience & Neuromorphic Engineering (CNET)

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# Questions